

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

**DIGITAL FOOD PLATFORMS AND HOUSEHOLD FOOD SECURITY:
UNDERSTANDING ACCESS, EFFECTIVENESS, AND BARRIERS**

An Undergraduate Thesis Presented to the
Faculty of the College of Arts and Sciences
Rizal Technological University

In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Statistics

by

Arvin H. Candelaria

Rosalyn N. Abalos

Vince Saiha T. Arcos

Dan Karlo M. Vullag

July, 2025

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

ii

APPROVAL SHEET

This thesis entitled “**Digital Food Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers**” prepared and submitted by **Arvin H. Candelaria, Rosalyn N. Abalos, Vince Saiha T. Arcos, and Dan Karlo M. Vullag** in partial fulfillment of the requirements for the degree of Bachelor of Science in Statistics has been examined and recommended for final and oral examination.

May 7, 2025
Date

Faustino E. Oguan, Jr. PhD
Adviser

Approved in partial fulfillment of the requirements for the degree of **Bachelor of Science in Statistics** by the Oral Examination Committee with a grade of _____.

Virginia S. Sobremisana, PhD.
Chairperson

Susan E. Puyat, PhD
Member

Mylene C. Prado, MA
Member

Accepted in partial fulfillment of the requirements for the degree of **Bachelor of Science in Statistics.**

Date

Jeremiah V. Pacer, DPA
Dean

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

iii

ACKNOWLEDGEMENTS

The researchers express their sincere gratitude to the following who supported the completion of this thesis:

They especially thank their adviser, **Dr. Faustino E. Oguan Jr.**, for his guidance, support, and valuable insights;

They also acknowledge the internal validators from Rizal Technological University – Boni Campus: **Dr. Omar R. Manlapas**, **Mr. Hans Christian F. Delos Reyes**, and **Dr. Lea S. Nisperos**, as well as the external validator from the Polytechnic University of the Philippines – Sta. Mesa Manila, **Dr. Elvira C. Labartine**, for their constructive feedback and contributions;

Appreciation is given to **colleagues and friends** for their encouragement, and to their families for their unwavering love and support; and above all, the researchers thank **God** for making the impossible possible.

A.H.C
R.N.A
V.S.T.A
D.K.M.V

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	iv
	ABSTRACT As digital technologies reshape everyday practices, online food delivery platforms have emerged as a significant avenue for food access in urban areas. This study explores the role of digital food platforms in household food security in Mandaluyong City, Philippines. Specifically, it examines how these platforms influence food accessibility, affordability, and reliability, while identifying barriers to their practical use. Given the urban context and increasing dependence on digital solutions, understanding these dynamics is crucial for developing inclusive and responsive food systems. Employing a descriptive-correlational research design, the study surveyed 400 respondents selected through stratified random sampling across key barangays. Quantitative data were collected using a validated questionnaire and analyzed using frequency, percentage, mean, standard deviation, independent samples t-tests, and one-way ANOVA. These statistical tools enabled the researchers to analyze patterns in food delivery usage and assess the influence of demographic and socioeconomic factors. Findings reveal that most respondents were young adults aged 18–34, predominantly male, from low-income households, and employed in working-class sectors. Despite widespread digital access, respondents used online food delivery platforms only occasionally. Most households reported no or rare instances of food insecurity, although mild food-related concerns persisted. Respondents expressed dissatisfaction with affordability, convenience, and food quality, while older and	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	v
	less tech-savvy users struggled with digital literacy and app usability. Demographic factors, including age, sex, income, and education, significantly influenced both usage patterns and perceived barriers. The study concludes that while digital food platforms offer potential benefits, their effectiveness remains limited by economic, technological, and trust-related constraints. These findings underscore the importance of targeted policy interventions and platform enhancements to improve food access and promote digital inclusion in urban communities.	

TABLE OF CONTENTS

	Page
Title Page	i
Approval Sheet	ii
Acknowledgements	iii
Abstract	iv
Table of Contents	v
List of Tables	viii
List of Figures	ix
CHAPTER I THE PROBLEM AND ITS BACKGROUND	
Introduction	1
Theoretical Framework	3
Conceptual Framework	5
Statements of the Problem	6
Hypotheses	8
Significance of the Study	8
Scope and Limitations	11
Definition of Terms	13
CHAPTER II REVIEW OF RELATED LITERATURE	
Global Food Insecurity in the Digital Age	18
Convenience and Accessibility of Food Technology Services	21

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

vii

Affordability of Food Technology Services	28
Technology-Driven Food Redistribution Programs	34
Technology-Driven Food Redistribution Programs Review Platforms and Apps that Connect Surplus Food Providers with Organizations or Individuals in Need such as Food Banks or Charities	37
Digital Transformation in Food Supply Chains	42
Barriers to Technological Adoption in Food Distribution Networks	45
Trust and E-commerce Acceptance in the Philippines	48
Ethical, Social Implications, and Role of Data Analytics in Tackling Hunger	56
The Role of Big Data in Addressing Hunger	56
Trust, Transparency, and Accountability in Data Analytics	57
Ethical and Social Implications of Data-Driven Food Systems	57
Distributed Ledger Technologies and Data Ethics	58
Policy Recommendations and the Future of Ethical Data Collaboration	58
Case Studies on Technology-Supported Food Distribution Systems	59
Geospatial Technologies in Food Distribution	63
Synthesis	65
CHAPTER III RESEARCH METHODOLOGY	
Research Method Used	68
Description of the Respondents	71

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

viii

Instruments Used	73
Data Gathering Procedures	75
Statistical Treatment of Data	76
CHAPTER IV PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA	
Demographic and Socioeconomic Profile	80
Extent of Reliance on Online Food Delivery	83
Household Food Insecurity Access Scale	84
Factors Influencing Use of Online Food Delivery	86
Common Barriers to Adoption	92
Demographics and Food Insecurity	98
Digital Delivery and Food Insecurity	100
CHAPTER V SUMMARY, CONCLUSION, AND RECOMMENDATIONS	
Summary of the Findings	102
Conclusion	106
Recommendations	108
References	111
Appendices	
Appendix A - Letter to the Respondents	120
Appendix B – Instrument used to the respondents	121
Appendix C - Letters to the Validators	123
Appendix D – Letters to the Barangays	127

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

ix

Appendix E – Letter to grammarian

131

Curriculum Vitae

132

LIST OF TABLES

Table	Page
1. Frequency and Percentage Distribution of the Respondents according to age	80
2. Frequency and Percentage Distribution of the Respondents according to sex	81
3. Frequency and Percentage Distribution According to employment status	81
4. Frequency and Percentage Distribution According to Income level	82
5. Frequency and Percentage Distribution According to Occupation	83
6. Frequency and Percentage of Online Food Delivery Consumption	83
7. Level of Food Insecurity	84
8. Convenience and Accessibility	86
9. Affordability and Cost Considerations	87
10. Digital Literacy and Payment Preferences	89
11. Food Quality and Availability	91
12. Financial Barriers	92

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

x

13. Digital Barriers and Technical Issues	93
14. Service and Reliability Concerns	95
15. Trust and Security Concerns	97
16. Demographic and Socio-Economic Profile and HFIAS	98
17. Differences in Factors Influencing Food Delivery Used by Demographics	100

LIST OF FIGURE

Figure		Page
1	Food Distribution Efficiency (Conceptual Framework)	5

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter discusses the nature and background of the study. The paper presents the introduction, theoretical framework, statements of the problem and hypotheses, the study’s significance, scope and delimitation, as well as the definition of terms.

Introduction

Digital platforms have significantly transformed food access and consumption patterns, particularly in urban areas like Mandaluyong City where digital connectivity is widespread. Online food delivery platforms have become indispensable for modern households providing convenience and diverse range of food options. However, the impact of these platforms on household food security raises important concerns.

Despite the increasing adoption of online food delivery platforms in urban settings such as Mandaluyong City, there remains a lack of empirical research on how these platforms influence household food security particularly among low-income and food-insecure households. Most existing studies focus on technological usability, platform efficiency, or consumer convenience (Yaiprasert & Hidayanto, 2024; Balgomera et al., 2022), yet there is a limited understanding of how these platforms intersect with structural barriers such as affordability, digital literacy, and socio-economic disparities (Capistrano, 2021; Lim, 2014).

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	2
	Furthermore, while some global and regional studies acknowledge the role of digital services in food access (Akinbamini et al., 2025; Olajide et al., 2024), there is a distinct lack of localized data and analysis specific to highly urbanized, densely populated areas in the Philippines, such as Mandaluyong City. Another gap lies in the evaluation of online food delivery platforms through the lens of food security frameworks—specifically, how accessibility, affordability, and usage patterns relate to the four dimensions of food security: availability, access, utilization, and stability. Few studies, if any, systematically assess whether these digital tools genuinely enhance food security outcomes or replicate existing inequalities in a digital format. Additionally, the existing literature frequently overlooks whether digital platforms significantly contribute to achieving Sustainable Development Goals (SDGs) such as reducing hunger (SDG 2), promoting well-being (SDG 3), or enabling responsible consumption (SDG 12). This gap in research raises concerns about whether these platforms are genuinely equitable and sustainable solutions or if they might exclude the populations that are most in need. This study addresses these gaps by investigating the real-world experiences of households in Mandaluyong City, examining the complex relationship between digital food delivery services and food security, with an emphasis on marginalized communities. The research aims to fill this void by offering contextual, evidence-based insights that can guide inclusive digital food recommendations.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	3
	Theoretical Framework The Technology Acceptance Model (TAM) by Davis (1989) and Jack Gibb’s Theory of Reasoned Action (TORI), Trust Theory (1972) can be connected through their influence on the adoption and effectiveness of technological advancements in food distribution systems. The Technology Acceptance Model (TAM) posits that two primary factors influence an individual’s acceptance of technology: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). Perceived Usefulness (PU) refers to the degree to which a person believes using a specific system will improve their performance or efficiency. In the context of food distribution platforms, consumers are more likely to adopt these technologies if they perceive them as beneficial in ensuring food security, reducing waste, and enhancing accessibility. Perceived Ease of Use (PEOU) represents the perceived ease of use of a technology. Users who find digital food delivery platforms intuitive and convenient are more likely to integrate them into their daily routines. Furthermore, Behavioral Intention to Use (BIU) is influenced by both Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), meaning that the higher the perceived usefulness and ease of use, the greater the likelihood of adoption. Trust Theory emphasizes that high trust in relationships-with family, friends, or institutions—enhances cooperation and engagement. In the context of food delivery apps, users who trust the technology and the companies providing these services are more likely to utilize the platforms regularly. Trust can develop	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	4
	through positive experiences, transparency in food sourcing and safety, and clear, reliable communication. When these two theories intersect, trust becomes evident as essential for the successful adoption of digital food distribution platforms. Even if a technology is perceived as valuable and easy to use, low trust can still deter potential users from engaging with it. Conversely, a high level of trust can mitigate concerns and encourage wider adoption. Strategies to foster trust—such as secure payment systems, transparent food safety practices, and user-friendly interfaces—can significantly improve engagement with digital food platforms. This study examines how digital food platforms can enhance food security and distribution efficiency while addressing adoption barriers, particularly in underserved urban areas. By integrating TAM and Gibb’s Trust Theory, the study establishes a strong theoretical foundation for assessing the real-world application of technology in food distribution. TAM helps structure the analysis of how users perceive digital food delivery technologies, while Trust Theory highlights the role of confidence in service reliability, transparency, and safety. These theories offer a comprehensive perspective on the behavioral and systemic challenges of technology adoption. This framework enables researchers to identify key factors that influence the adoption of digital food platforms, ultimately guiding the development of more effective and trustworthy food distribution networks to combat hunger.	

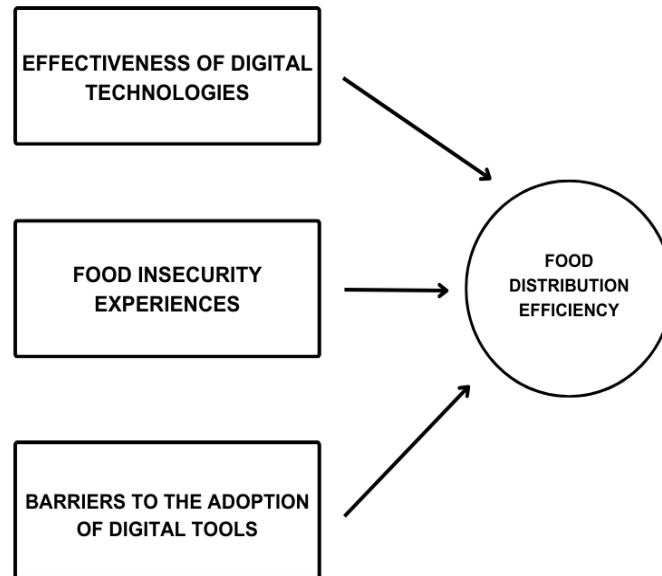
Conceptual Framework*Figure 1*

Figure 1 presents the conceptual model of the study titled "Digital Food Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers". This model illustrates the interconnected relationship between digital technologies, food insecurity experiences, and the barriers to technology adoption, all of which influence the overall efficiency of food distribution systems in urban settings. Digital tools such as food delivery applications and cashless payment systems like GCash, credit cards, and bank transfers are seen as enablers of improved food distribution. Their effectiveness depends on how well they enhance accessibility, streamline supply chains, reduce food waste, and provide convenient access to food services. However, food insecurity remains a persistent challenge, especially in underserved areas, where issues of

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	6
	affordability, inconsistent availability, and logistical inefficiencies impact how communities interact with these platforms. The model also highlights critical barriers that limit the adoption of digital tools, including poor internet connectivity, a lack of digital literacy, mistrust in online transactions, and concerns about data security. They must address these barriers to ensure that digital food platforms effectively serve all population segments. Ultimately, the model positions food distribution efficiency as the central outcome, determined by the successful integration and acceptance of digital technologies, especially among vulnerable and low-income groups. Overcoming these challenges can result in a more inclusive, equitable, and sustainable urban food system. Statements of the Problem This study aims to examine the role of online food delivery platforms in household food security in Mandaluyong City, focusing on their accessibility, affordability, effectiveness, and barriers to adoption. Specifically, it seeks to answer the following questions: 1. What is the demographic and socio-economic profile of households in Mandaluyong City in terms of: 1.1. age; 1.2. sex; 1.3. employment status; 1.4. household income; and	

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

7

1.5. occupation?

2. What is the frequency of online food delivery received by the respondents?

3. How often do the respondents experience food insecurity?

4. What are the factors influencing Online Food Delivery Usage in terms of:

4.1. convenience and accessibility;

4.2. affordability and cost considerations;

4.3. digital literacy and payment preferences; and

4.4. food quality and availability?

5. What are the common barriers to adopting online food delivery platforms in terms of:

5.1. financial barriers;

5.2. digital barriers and technical issues;

5.3. service and reliability concerns; and

5.4. trust and security concerns

6. Is there a significant difference on the factors influencing the respondent's use of online food delivery when grouped by age, sex, employment status, household income, and occupation?

7. Is there a significant difference on the barriers to respondent's adoption of online food delivery platforms when grouped by age, sex, employment status, household income, and occupation?

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	8
	Hypotheses The study “Digital Food Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City” is used to reject the following hypotheses: H₀₁: There is no significant difference on the factors influencing the respondent’s use of online food delivery when grouped by age, sex, employment status, household income, and occupation. H₀₂: There is no significant difference on the barriers to respondent’s adoption of online food delivery platforms when grouped by age, sex, employment status, household income, and occupation. Significance of the Study This study not only helps people understand specific demographic behaviors but also guides them on how to understand food platforms more effectively to achieve food security and alleviate hunger. The beneficiaries of this study include: Low-income Households and Food-Insecure Families: This study is significant for low-income households and Food-Insecure Families as it provides a deeper understanding of the challenges they face, particularly in terms of poverty and limited access to adequate nutrition. By identifying the underlying social and economic factors that contribute to food insecurity, the research shifts the discourse from short-term aid to long-term, systemic solutions. It serves as a	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	9
	basis for designing more effective and sustainable interventions that promote food security, social equity, and improved quality of life for vulnerable populations. Furthermore, the study on Low-income Households and Food-Insecure Families contributes to academic discourse by enriching the body of knowledge on urban poverty, food access, and the role of policy and community-based responses in addressing these pressing issues. Due to the prevalence of low-income households and food insecurity, they will also benefit from the development of targeted programs and policies that are responsive to the actual needs of food-insecure households, paving the way for more inclusive and impactful approaches to poverty alleviation and nutritional well-being. Digitally Illiterate or technologically disadvantaged individuals: This study highlights the exclusion of digitally illiterate or technologically challenged individuals from digital food platforms, revealing inequalities that hinder access to essential foods. Raising awareness of these issues uncovers barriers that significantly impact household food security for vulnerable populations. Addressing these gaps can lead to more inclusive policies and innovations, making digital platforms more effective. Improved access to food assistance, community-focused platform designs, and stronger support systems can ensure that everyone participates in the digital era. Recognizing these challenges allows for solutions that enhance food security at the household level. Closing these gaps will promote the development of inclusive policies and innovations, ensuring that in the digital age, no one overlooks anyone.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	10
	Policymakers: Policymakers in Mandaluyong City can rely on this study to develop smarter and more inclusive ways to ensure families have access to food through digital platforms. Understanding who can and cannot access these platforms allows them to identify gaps in internet availability and digital skills, ensuring that no one is left behind. Evaluating the effectiveness of these platforms in helping people obtain food can lead to improvements in existing programs or the creation of new tech-driven solutions for food distribution, price monitoring, and emergency food aid. Addressing challenges such as high costs, limited internet access, or difficulties in using these platforms enables leaders to implement real solutions, including affordable internet, digital skills training, and partnerships with businesses to enhance food accessibility. Additionally, these insights can help shape policies that reduce hunger, strengthen food security, and build a more resilient and adaptive food system for everyone in the city. Future Developers: The researchers envision opportunities for developers to create a single application that consolidates food delivery services. Additional platforms and apps, like TikTok, Shopee, and Messenger, could also facilitate food sales. The proposed app should include a pre-approval or pre-registration process for online food sellers and small-scale vendors, allowing them to submit applications based on their market location to enhance food delivery accessibility and ensure product quality standards. When users place orders through this app, the system should automatically track the food's location. Similar to shopping platforms like Shopee and TikTok, purchases could earn points. This points	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	11
	system is inspired by Ecosia's model, where users contribute to tree-planting initiatives by using the search platform. In contrast, the proposed app would function as a food bank aimed at reducing food waste and combating hunger. Future Researchers: This study lays a solid foundation for future researchers examining the dynamics of the interface between digital technology and food security, particularly among individuals who are digitally illiterate or technologically challenged. The contents of this study shed light on the difficulties these groups face with digital food platforms, and therefore also highlight some neglected barriers affecting food access and equity. Understanding such factors motivates further research, which, in turn, leads to more inclusive technological solutions. Potential benefits include user-friendly platforms and enhanced digital literacy programs and policies that ensure food distribution systems are accessible to all, regardless of their technological ability. Scope and limitations of the study This study covers several important topics. The demographic and socioeconomic characteristics of online meal delivery consumers are first examined, emphasizing how factors such as age, gender, income level, job status, or occupations influence the adoption of food delivery platforms. To determine if households rely on food delivery platforms for their regular meals or use them sometimes, the study also looks into the degree of household reliance on these platforms. Third, it assesses how online meal delivery platforms affect household	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	12
	food security, determining if they impose more financial burdens or increase food access. Fourth, the study examines the factors—such as cost, convenience, digital abilities, and individual dietary preferences—that affect the adoption of these platforms. Finally, it examines the obstacles to the uptake of online meal delivery, including budgetary limitations, illiteracy, restricted payment choices, and accessibility issues for lower-income households. Despite its thorough approach, this study has several limitations that need to be noted. First, because the study only looked at Mandaluyong City, its conclusions might not hold for other cities or rural regions with different access and infrastructure for food delivery. The study also uses self-reported data from questionnaires, which can add biases, inaccuracies, or recollection problems because participants might not always give accurate or impartial answers. The surveys were intended to collect numerous demographic and socioeconomic data points, and statistical techniques were used to examine the data to find patterns and connections; however, researchers failed to collect the household size, which can also determine food insecurity, and inadvertently added to data the primary method of online delivery platforms and occupation, which the researchers then think would have less impact to the overall result. Thirdly, the study only examines a single period and ignores long-term patterns or shifts in the economy that can affect the use of food delivery platforms and households' food security. Fourth, the study ignores other initiatives addressing food insecurity, such as traditional food markets, food banks, and government aid programs, in	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	13
	favor of concentrating solely on digital food delivery platforms. Lastly, despite their importance to the topic, households without cell phones or internet access may be underrepresented in the findings due to technological constraints. Despite their difficulties, these limitations do not lessen the importance of this study. Instead, they draw attention to areas that need more research, like long-term trends in food security, comparative studies with other cities, and more comprehensive assessments of digital and non-digital food access options. Policymakers and other stakeholders can use this study's insightful analysis of the relationship between food security and digital technologies to improve the affordability and accessibility of food for disadvantaged groups. By understanding how online meal delivery platforms impact household food security, this study contributes to the broader discussion on food availability. It gives optimism for a time when technology can close socioeconomic divides. Definition of Terms Affordability. A consumer's financial capacity to purchase a good or service without endangering overall economic stability. Analysts often gauge this by comparing the price against monthly earnings, then assessing whether the outlay still allows for the payment of rent, utilities, groceries, and other essential expenses. Availability. The actual readiness of a product for immediate use or acquisition when a need arises. The concept encompasses whether a brick-and-	

mortar store displays the item, whether a digital vendor can dispatch it without delay, and whether strikes, floods, or other disruptions in the supply chain might suddenly render the supply elusive.

Convenience. Speaks to how quickly and smoothly someone can carry out a task in daily life. For food delivery apps, convenience is evident when a person taps a few buttons, watches the progress on a map, and greets the rider at the door once the meal arrives—no password resets, dead links, or surprise hidden fees involved.

Cost Considerations. Gather all the money-related questions that come to mind before making a purchase. Shopping is not just about the sticker price; it is also about how that price sits inside the household budget and whether the time saved, the taste promised, or the bonus rewards make the expense feel worthwhile.

Digital Barriers. Obstacles that prevent people from entirely using the digital tools at hand. They often arise from outdated devices, slow or spotty Internet service, or someone feeling shaky about basic tech skills.

Digital Literacy. The ability to effectively use technology, including food delivery applications, for ordering and payment. A lack of digital literacy can hinder accessing food through digital platforms.

Financial Barriers. Day-to-day money woes that stop people from buying groceries, school supplies, or a bus fare. They may spring from sky-high prices,

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	15
	wages that have barely moved, or must-pay bills that swallow the small slice of income originally set aside for essentials. Food Access. The ability of individuals to obtain food, determined by financial resources, food prices, and market accessibility. Socio-economic conditions are key in whether people can afford and reach available food sources. Food Availability. Refers to the physical presence of food in a specific area. This availability relies on several factors, including agricultural production, food storage, distribution, and trade. It ensures that food supplies are sufficient for the population. Food Insecurity. Refers to when individuals or households lack reliable access to nutritious food due to financial, social, or physical constraints. It ranges from mild concerns about food access to severe hunger and malnutrition. Food Quality. The mix of traits that shapes a shopper's view of food, including nutrition, freshness, smell, taste, and sheer pleasure. It also includes whether the product meets hygiene standards, protecting customers' health while adding real value to the shopping experience. Food Security. The ability of individuals, households, and communities to consistently access sufficient, safe, and nutritious food to meet their dietary needs for a healthy and active life. It is influenced by economic stability, social equity, and agricultural sustainability (FAO, 1996).	

Food Stability. The consistency of food availability and access over time. Stability is affected by economic fluctuations, climate change, and political instability, which may disrupt food supply chains.

Food Utilization. Refers to how food is consumed and absorbed by the body, influenced by dietary diversity, food safety, and individual health conditions. Proper utilization ensures nutritional well-being.

Household Food Insecurity Access Scale (HFIAS). A standardized tool developed to measure the level of food insecurity within a household, assessing concerns about food availability, meal quantity, and dietary diversity.

Online Food Delivery Platforms. Digital platforms such as GrabFood and FoodPanda allow consumers to order meals from restaurants and deliver them to their homes. These services influence food accessibility, affordability, and convenience.

Payment Preferences. The exact way a customer chooses to settle up when buying online. Options usually hinge on comfort, perceived safety, and the menu available, ranging from debit and credit cards to prepaid vouchers, contactless mobile wallets, or the traditional cash-on-delivery method.

Service and Reliability. Covers how consistently and dependably a provider meets its promises to customers. Key indicators include on-time delivery, accurate order assembly, responsive technical support, and a consistently high level of quality across successive transactions.

Socio-economic Factors. Refer to elements like income level, employment status, and household size that affect a household's capacity to access food and utilize online food delivery services.

Technical Issues. The label encompasses the entire list of problems that arise the instant a laptop, software, or network refuse to behave—a forgotten line of code, a frayed cable, or a shaky Wi-Fi signal. Just one hidden glitch can disrupt a person's daily routine and turn the most mundane task into a lengthy diversion.

Trust and Security. This measure indicates how comfortable a visitor is that a site will keep private data and credit card numbers secure and out of prying eyes. Strong shields against hackers, together with clear, regular updates on those defenses, boost that comfort and offer users fresh reasons to come back.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter provides a comprehensive review of the existing literature on technology use in food distribution networks. By analyzing global trends, technological advancements, and local challenges, the evaluation aims to contextualize the study and identify gaps in current research that this study seeks to address.

Global Food Insecurity in the Digital Age

The world's population is growing steadily. Thus, initiatives and technology to boost agricultural production sustainably, reduce food waste, and enhance the global supply chain are desperately needed to guarantee the availability of wholesome food. Nearly 828 million people suffer from hunger, and by 2030, about 670 million people or 8% of the world's population will still be undernourished, according to the United Nations State of Food Security and Nutrition in the World 2022 study. The Green Revolution supported the expanding population and lessened a nation's import need for several decades.

However, it negatively impacted biodiversity and deteriorated soil (Sharma et al., 2024). In order to ensure that wholesome food reaches the masses, tackling food insecurity will improve distribution since it highlights the necessity of sustainable technologies that increase agricultural output, decrease food waste, and optimize supply chains (Sharma et al., 2024). These tools can help make agriculture more efficient, reduce environmental impacts, and improve

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	19
	resilience against climate change and natural disasters. The integration of sustainable practices and digital solutions is key to creating a food system that is both productive and equitable, which can help alleviate food insecurity and enhance distribution efficiency. Based on the study of Yaiprasert and Hidayanto (2024), to fill a significant knowledge vacuum regarding AI-based decision-making in the restaurant and food delivery sectors, the study introduces and evaluates a unique strategy that uses ensemble machine learning approaches for tailored meal offerings. Ensemble machine learning algorithms and AI-powered decision-making suggest a framework that may enhance food accessibility, particularly for people with specific dietary requirements. The study's emphasis on individualized and nutrient-dense meal recommendations underscores the significance and potential impact of the findings. These recommendations have broader implications for improving food distribution by offering customized options encouraging better lifestyles, even though they do not directly address widespread food insecurity. Food systems may benefit from investigating AI's ability to create new employment opportunities, particularly in underprivileged areas. Finding services like food banks or aid programs is more difficult for people experiencing food insecurity because not everyone can access technology. Many food insecure people, particularly those who reside in rural or low-income areas, might not have access to the internet or be digitally literate, which would limit their capacity to	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	20
	use financial aid applications or online food delivery services. The digital divide must be closed so that technological breakthroughs in food distribution can reach those who need them most. Globally, the percentage of older persons is rising, but many of them face food insecurity. Technological advancements like easier access to food delivery apps (FDAs) on smartphones and the internet may lessen food insecurity, but not enough research has been done on this subject. In addition to highlighting equity issues, such as the difficulties experienced by older persons in rural regions or those with physical impairments, Hwang et al. (2024) address the possible advantages of FDAs in enhancing food access. The intricacies of utilizing technology in food delivery are further shown in the emphasis on personalized app features to boost usage and evaluate health and environmental effects. The statement does concede, nonetheless, that additional study is required to comprehend the FDA's capacity to remedy food inequality completely. Additionally, it recognizes that customized app features tailored to the specific needs of older adults, such as user-friendly interfaces and additional support functionalities, can increase the likelihood of FDAs being used. This indicates that when thoughtfully designed, technology can significantly impact food distribution, offering more efficient and convenient ways to access nutritious meals, especially for populations struggling with traditional food procurement methods.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	21
	Furthermore, it explores the broader implications of FDA usage, such as its potential health and environmental impacts, including food waste and the promotion of ultra-processed foods, a critical consideration for sustainable and equitable food systems. This proves that during COVID, the hunger trend rose to 270 million severely food-insecure people and continued to bear malnutrition (Podkolzina et al., 2023). The food shortage situation is still critical for many countries after COVID. Vedovato et al. (2022) highlight that in small communities, low-income families provide important insights into the various factors that influence their access to healthy food and grocery shopping. Their perspectives are vital for developing policies that effectively meet the needs of underserved communities and promote equity in healthy eating. Incorporating family perspectives into the formulation of food policies is a potent participatory strategy for empowering and enhancing community capacity. Insufficient research on disparities in food access includes community perspectives to understand better-reported problems and the direct living circumstances of low-income families. Lee et al. (2019) support and align it with the Theory Acceptance Model (TAM), which has been used to explain users' behavioral intents, and the UTAUT model, which broadens the research on food services and consumer behaviors. Convenience and Accessibility of Food Technology Services New research in food technology emphasizes the growing influence of convenience on consumer choices. In this case, Garrido and Gallardo (2022) posit that consumer adoption of new food processing technologies is primarily	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	22
	convenience-based. As to the general level of access to food services, research Garrido and Gallardo did shows that consumers do enjoy the convenience of technological improvements, but convenience alone does not guarantee widespread acceptance. This study found that the consumer in question takes into consideration not only the time savings provided but also the quality of the service and its impact on health. The authors draw special attention to the lack of adequate education and trust, which affects the level of acceptance of food tech innovations. These findings help explain why convenience, in this case, is restricted so much when factors such as disclosure and perceived worth are absent. This illustrates the need to consider many consumer issues in order to boost adoption of technology enhanced food systems, especially in the context of digitized and processed foods. Equitable access enabled through digital means is an emerging focus area to consider in food technology services. The works of Kalinowska et al. (2021) show that mobile technology, for example, plays a great role in food quality evaluation access through smartphones. The study shows that mobile sensors and applications enable people to make informed choice about food in real time, especially in urban areas. In addition, the study emphasizes how economical and simplicity of these tools' design makes them available to a larger section of the population. Kalinowska et al. substantiate these claims by demonstrating that mobile technology-based tools undermines the information inequality concerning food safety regarding the public to promote health equity. The drawback of low	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	23
	skills needed to operate a mobile device disqualifies the device accessibility assumption from many social economic classes. All these factors contribute to this study on low cost food technology services with great emphasis on how mobile devices can transform the food landscape boost accessibility information inequality fracture zone within food environments. According to the research of Fernandez and Raine (2021), there is clear potential for the enhancement of nutrition access and reach of food services through digital food retail which is the context of public health. It was noted in this study that online grocery shopping has the potential to increase food availability for various persons on a wider scope, particularly for those who live in far-flung or underserved regions. Additionally, the authors highlight the role of public policy as well as health communication in determining the efficiency of digital food services. Fernandez and Raine showed that healthy food can increasingly be made available at lower prices and greater convenience through digital platforms, which is consistent with other research that underscores the importance of including people digitally. However, they also point out concerns regarding data privacy and the commercialization of health messages. These findings provide a foundation for understanding how digital food services can serve public health goals, reinforcing the need for responsible and inclusive technology design in food access initiatives. Perhaps one of the most prominent trends in food consumption is the increasing attention paid to organic food and the use of digital platforms in its marketing.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	24
	Kamboj et al. (2023) state that the ease of access and the transparency of the information are fundamental determinants encouraging consumers to purchase organic products online. The study found that digital platforms facilitate easier access by minimizing the work needed to locate and compare alternatives of organic foods. Moreover, the real-time data and other relevant information, customer testimonials, and automated cash registers bolster trust and the intention to purchase. Cost and doubts regarding the legitimacy of the certification are still limitations, however. These findings serve to illustrate how digital convenience shaped consumers' dietary choices, which, in this case, promotes health. This study aims to show the need to remove credibility barriers to increase the availability of reliable organic food services. As noticed by Sgroi (2023), a new trend in the digital food environment is the combination of culinary heritage with information technology. This study found that food technology services can help preserve traditional cuisines and make them available to contemporary consumers. Moreover, the research emphasizes the behavioral economics dimension of food choice by observing that cultural identity as well as familiarity affect platform usage. Even though these digital initiatives broaden access, opportunities related to ensuring authenticity and community participation are still challenges. These findings are pertinent to the interest of this study in that they illustrate culture must be integrated to true accessibility and sustainability in technology infrastructures. In food tech services,	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	25
	these outcomes call for a reexamination of the way’s innovation is done relative to traditional elements. This is the fourth essay in a sequence: the last sentence in the previous subsection draws attention to “digital spaces.” From “virtual worlds,” to digital food environments and online ecosystems, the expositions have revolved around geography and sociography, within which people interact with places. In this regard, recent investigations regarding food environments have focused on how accessibility increases for mitigating health risks like obesity. According to Eskandari et al. (2022), both the analog and physical elements of the food environment greatly dictate one’s dietary habits as well as availability of options to choose from. Within the framework of food technology services, their synthesis of different study designs showed that food insecurity and limited access to healthy foods remains a primary barrier to attaining well-being even within high-income countries. This study showed that obesity mitigations could be achieved through public health initiatives via digital service platforms that provide healthier food options, particularly when these initiatives are synergized with public health campaigns. Also, the results highlight the value of digital delivery systems and applications in the closure of these gaps. This study, however, demonstrates who stands to gain from these services, which is determined by socioeconomic standing. Moreover, these results contribute to the discourse regarding access to	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	26
	food technology resources and inadequate assurances from makers and distributors that are meant to promote health. These days, the digital food setting is getting more and more of an attention in public health research. Granheim et al. (2022) explored the intersection of mapping technologies and food services by undertaking a systematic scoping review of digital food environments. Their study pointed out the role of online platforms in determining food selections and improving access to multiple food services. In addition, it shows that custom-made algorithmic suggestions as well as keyed advertisements influence eating habits. Granheim et al. show that better access to food can be achieved in a digitally-enabled environment by integration of nutritional instructions and other health-relevant information. Similarly, Fernandez and Raine (2021) support public health concern from digital infrastructures. On the other hand, absence of standard policies regarding digital food environments is still applicable. These findings seem to resonate with the emphasis of this study by arguing that a systematic design approach that prioritizes users in digital food solutions can increase both accessibility and better nutrition. Within the scope of area studies, Wu et al. (2022) explain how cultural and economic considerations impact the use of food delivery applications (FDAs) in Asia and the Philippines specifically. This study noted that convenience, time-saving, and promotional constraints greatly affect users' decisions to access FDAs.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	27
	Furthermore, the studies highlight how the younger customers are facilitated an easier navigation and payment system, and how these features positively impact adoption. Chawla and Puri (2022) also show that the issues of safety and dependability of services have a major influence on the constituents' selection decisions. This was also discussed by Garrido and Gallardo (2022). In any case, the issues resulting from the digital divide still limit the wider utilization of the services by older or less technologically inclined individuals. These studies help explain the local phenomena associated with the adoption of digital food services while accentuating the need for fostering adaptable frameworks and cultural contexts. As with all useful innovations on a sustainable food system, the integration of food access enhancing technology with climate-resilient crops strategy is an epitome of a milestone. Noort et al. (2022) argue that the inclusion of local food crops into the digital marketing systems of foods enhances both accessibility and nutrition. This study found that bread-like products made from ethnic staples had a higher acceptance rate among online shoppers when marketed as sustainable. Moreover, the study illustrates the way health- and eco-sensitive consumers can access producers through digital means. Noort et al. demonstrate that food technology services can catalyze dietary changes when these services emphasize the effects on health and the environment. But, there are still some supply chain issues and conflicting narratives which pose challenges. These findings are of particular interest to this study's aims because they claim that gaps	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	28
	in the provision of sustainable food using digital means can be closed by proper communication strategies and systematic supply. This Chotigo & Kadono (2021) study analyzes the acceptance of digital food delivery platforms during the COVID-19 pandemic and their integration within consumer lifestyles. The study argues that perceptions of safety, ease, and time efficiency (specially the possibility to save “time” by avoiding traffic jams or long queues) significantly impacted the use of food delivery apps during the pandemic period. Moreover, it observes that previous concerns about expenses and being “too old” technologically shifted to concerns about speed and exposure risk. Even the most reluctant users started to use FDAs when it became essential for personal safety, a fact that is corroborated by findings of Pandey et al. (2022). However, these authors emphasize that the retention of such users as long-term users, especially with the gradual return to in-person dining, is highly questionable, though. These studies underscore the idea that crises can foster the rapid uptake of digitized services, while at the same time illustrating the paradox of the food service platforms’ struggle to maintain dependability and trust. Affordability of Food Technology Services Recent studies on food technology highlight the growing significance of affordability in technology use. According to Towse and Mauskopf (2018), affordability is one of the biggest challenges when introducing new technologies, especially in food systems. In the context of food tech, their research shows that new innovations might not reach low-income users if the cost is too high. This	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	29
	study identified that even when a technology is helpful or improves quality, if it is not affordable, it will not be widely accepted. Furthermore, they emphasize that balancing price and innovation is necessary for making technologies useful for all. These findings provide a foundation for understanding that cost must be considered when promoting food technology, especially in low- to middle-income settings. An emerging attention in nutrition field research is the growing concern with diet-cost effectiveness as one of the indicators of food system sustainability. Chungchunlam and Moughan (2024) found that already having a sufficient income, the cost of healthy diet is out of reach for a considerable number of people, particularly within the impoverished population. This study found that, even when people can access nutritious foods, lack of financial resources significantly impacts their ability to maintain a healthy diet. Furthermore, they illustrate that as a fundamental goal, sustainable food systems must include affordability. Herforth et al. (2020) also argues how diet's price is considered as a relevant barrier to accessing food. The findings of this research have particular relevance to food technology studies illustrating that policy to improve the affordability of food is necessary to improve health and sustainability impacts. Food price data on a global scale, as well as the costs associated with meal preparation at the individual level, has been studied extensively by Masters et al. (2023). Their study found that in most underserved	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	30
	regions, the cost of well-balanced and uncomplicated meals is economically unattainable for a significant proportion of the population. In addition, they point out that food technologies should aim to lower these costs by improving production, packaging, and distribution. Masters et al. highlight that even simple meals can be pricey if affordable pricing for food is not considered. This agrees with Chungchunlam and Moughan (2024) who place emphasis on the cost of a balanced diet. Yet, this research also points out a shortcoming: calculating cost is not enough; it needs to be coupled with quality and access. Those findings reaffirm the notion that food technology needs to shift its focus toward developing strategies for reducing the price of nutritious meals at a global level. According to Lee et al. (2013) affordability is a key indicator when tracking global food and diet trends. This study identified that price monitoring can help governments and organizations decide how to improve diet accessibility. Furthermore, their global study shows that poor people spend more on unhealthy food simply because it's cheaper. Lee and colleagues provide evidence that food technologies can help reduce costs if combined with policies that promote affordability. This aligns with Masters et al. (2023), who also show that diet cost affects food choice. However, Lee et al. also point out that even good technology will not help if pricing policies do not support low-income consumers. These findings provide a foundation for building strategies that combine food innovation with cost-focused approaches to improve healthy food access.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	31
	Recent studies on food spending behavior highlight the growing impact of income on diet quality. Kotykova et al. (2020) studied how Ukrainian households manage food consumption depending on how affordable food is. This study identified that when food becomes more expensive, people buy less or switch to lower-quality options. Furthermore, their research shows that affordability directly affects both food quantity and quality, especially for low-income families. This aligns with Towse and Mauskopf (2018), who stress that affordability must be part of any new food technology strategy. However, Kotykova et al. also point out that inflation and economic instability make it harder to keep food prices fair. These findings are especially useful for food tech researchers and developers aiming to create products that remain accessible despite changes in the economy. Under food accessibility and diet maintenance, Herforth et al. (2020) shows that the cost of healthy eating is accessible to very few people across the world, whether at the individual, household, or country level. This study highlighted that the concept of affordability goes beyond the price of food; it includes earnings and the local food economy’s infrastructure influence on the available choices. They further concluded that, in a lot of regions, even where healthy food is offered, it is not economically accessible to the majority. This also supports Chungchunlam and Moughan (2024) who provide evidence that cost is a predominant driver in sustainable food systems.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	32
	Nonetheless, Herforth et al. stresses that lack of local knowledge is critical because different regions, even within the same country, can order differently in terms of affordability. These conclusions highlight the efficiencies of utilizing food technologies to counter the disparity between attaining food security and diet quality. In food research, there has been greater focus on the issue of good quality and reasonably priced food services. Like Piližota (2012) states, it is clear that people desire food which is healthy and affordable. This study pointed out that price, especially for those with lower socio-economic status, limits their ability to purchase better quality goods. It also explains how marketers need to determine what people are willing to spend and design products which satisfy those needs. In the same way, Kotykova et al (2020) claim that there's a tendency to lower the quality of food offered when the price goes up. Nonetheless, Piližota notes that good value food always needs to be safe to eat and fulfill basic nutritional needs. This evidence supports the greater need of designing food technologies concentrating on price and quality intertwined with the consumer market. Igwe et al. (2024) emphasized managing smart sustainable supply chains to enhance food pricing, capturing the importance of its integration towards socio-economic improvement. They found that cutting down on food distribution and production costs greatly aids in making food affordable, especially in areas with price shifts and supply challenges. Furthermore, they demonstrate that technologies related to storage, transport, and inventory monitoring can improve price stability. This also supports	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	33
	Masters et al. (2023), who further emphasize funds needed for saves on the preparation of food. But Igwe et al. argue that prices might still increase due to high energy or import costs, even with strong supply chains. These results are quite important regarding this study’s focus showing how technology on the supply-side can tremendously reduce costs toward food affordability. Recent research on food innovations highlights the need to balance expenses with nutrition, health, and sustainability. As noted by National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Food and Nutrition Board; Food Forum, & Nicholson, A. (Eds.). (2022), novel food technologies, particularly alternative proteins, must be priced reasonably for broad public acceptance. The research found that while some innovations provide economic and environmental benefits, they are almost always overpriced. Moreover, the authors argue that there is a need for developers and policymakers to collaborate to enhance the accessibility of these products. This is consistent with Towse and Mauskopf (2018) who also argue that affordability serves as a primary barrier. The study does, however, point out the possibility of overly reducing costs compromising product quality. These findings highlight the need to look into the specific food tech price strategy gaps for new offerings. Research on estimating diet costs has recently gained interest. It is noted in the study by Headey et al. (2024), that measuring ‘affordability’ in a particular way could influence understanding of the food access problem. This study showed that by relying on some methods, healthy diets appear to be more	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	34
	affordable than other methods. Also, these authors argue that better estimation methodologies are needed which consider household budgets and actual food prices. Herforth et al. (2020) also recommend stronger data to support affordability calculations and analysis. But, as Headey et al. warn, inadequate or inappropriate information can lead to poorly informed policy choices. These findings underscore the need to measure food security and affordability with the appropriate methodologies, especially technologies aimed at developing affordable food in other regions. Technology-Driven Food Redistribution Programs The United Nations' Sustainable Development Goals (SDGs) 12 and 2 are essential to creating a sustainable, hunger-free world. SDG 12 calls for sustainable food production and consumption, aiming to reduce food loss and waste, while SDG 2 is focused on achieving zero hunger. In this context, technology is pivotal in advancing both of these goals. Research has highlighted how technological innovations can significantly impact various stages of the food value chain, from production to consumption, to help reduce waste and improve food redistribution. One recent study (Grewal et al., 2024) proposed a framework for understanding how these technologies can help address food loss and waste by improving food production, distribution, and consumption practices. It outlines how technology can influence key stakeholders at each stage of the process, providing valuable insights for future research and development to reduce food waste and increase food production.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	35
	Ultimately, these technological solutions are vital in the global pursuit of achieving zero hunger. This teaching case delves into sustainable hospitality practices within the Asia-Pacific region, particularly in the hospitality and food service industries. With sustainability becoming an increasing priority in this sector, many businesses struggle to implement effective, sustainable practices independently. KITRO, a Swiss start-up specializing in waste management, sees an opportunity to expand into this market, offering solutions to help these businesses reduce food waste (Martin-Rios et al., 2020). Through this case, students can explore key topics such as sustainability, innovation, strategy, and hospitality management and understand how these elements work together to drive positive change in the industry. By blending research, real-world examples, and emerging technological solutions, this case provides an opportunity to learn how we can contribute to a more sustainable future while helping to tackle global hunger and waste. Researchers in China, the world's largest platform economy, examined the lack of food-sharing platforms in the country. Using a vignette-based qualitative approach, they conducted semi-structured interviews with 35 potential users and analyzed the data through thematic analysis to explore adoption intentions. The results reveal that key factors from existing technology acceptance theories—such as performance expectancy, effort expectancy, social influence, hedonic motivation, facilitating conditions, and price value—play a significant role in shaping adoption intentions. Beyond these, context-specific influences like trust	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	36
	in other users and the platform, concerns about "losing face" (mianzi), and safety concerns during the pandemic also emerged as crucial factors affecting users' willingness to engage with these platforms. According to Tian et al. (2025), by expanding the technology acceptance literature, this research enriches scholarly discussions on the effectiveness of technological innovations in promoting food supply chain sustainability. It introduces new considerations related to platform service content and operating context, offering valuable insights for future development. The concept of waste regimes explores the global connections that contribute to food waste, offering a deeper understanding of how waste circulates within economies. By framing waste as a social relationship, the concept underscores that, alongside the circulation of value, waste also circulates, with both systems tightly interwoven. The author challenges commonly used metaphors in food and agriculture scholarship, such as value and supply chains, arguing that these frameworks fail to capture the complexity of waste's role within the system. According to the research by Gille (2012), the empirical data from both the Global North and South make it clear that the unequal distribution of uncertainty plays a crucial role in driving food waste in both regions. The connection between risk and waste extends beyond geographical and scale boundaries, suggesting that focusing solely on technological innovation or addressing food waste in specific countries or regions will not provide a comprehensive solution. These limited approaches are likely to deepen existing inequalities and overlook the root	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	37
	causes of food waste on a global level. Food waste (FW) in developing countries is becoming a significant obstacle to sustainable development and the establishment of effective waste management systems. These countries often struggle with environmental and sanitary issues due to incomplete or ineffective FW management practices. Differences in food waste generation between developing and developed nations show that income levels, population growth, and public participation in waste management are key factors influencing the situation. Given these challenges, the discussion explores recycling efforts, relevant regulations, and current food waste treatment technologies in developing countries, using case studies to illustrate key points. Taiwan is often pointed to as a successful example of effective food waste management and is presented as a model that other developing countries could follow. The conclusion suggests adopting an integrative management system as a more practical approach to tackling food waste in these regions (Thi et al., 2015). Technology-Driven Food Redistribution Programs Review Platforms and Apps that Connect Surplus Food Providers with Organizations or Individuals in Need such as Food Banks or Charities Current technology-facilitated food distribution studies exhibit heightened importance in utilizing digital platforms to curb student hunger. Olajide et al. (2024) report that technology-facilitated interventions for student hunger have become imperative in effectively managing food surplus. Their work confirms that	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	38
	integrating mobile and AI platforms maximizes food availability in community food security. In such work, digital tracking platforms are used to improve food distribution efficiency and loss, with automation confirming to maximize efficiency and loss at a minimum level. Besides, their work confirms that ease of use and community acceptance platforms maximize successful integration. Olajide et al. (2024) demonstrate intelligent algorithms improving food distribution efficiency, concurring with similar studies confirming that technology maximizes distribution efficiency. Despite this, infrastructure gaps and low digital literacy hinder its use at a more significant level. In providing a platform for technological dimensions in food distribution, such studies form a platform for researching technology in food distribution, a dominant theme in such work. Bridging infrastructure and literacy gaps maximizes efficiency in food access through technology. The impact of such a work confirms its necessity in researching user-facilitated designs to improve efficiency in localized communities. A key development in food reallocation programs is increased concern with integrating technology and health in urban food security resolution. As argued by the NYC Food Policy Center (Hunter College) (2024), technological innovation aids in providing urban communities with food by connecting food surplus providers with poor communities. The study reveals the effective use of information in decision-making in food reallocation improvement. Online platforms ease food donation logistics, particularly in urban settings where the	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	39
	availability and scarcity of food coexist. As argued in the study, technology in policy creation strengthens food security interventions. Regulatory barriers and uncertain funding, nevertheless, hinder such programs' expansion. This study reveals a need for integrating technological development with health policies to improve food reallocations. Overcoming such barriers regarding regulation and securing long-term funding could make such programs even more efficient. Technology integration in public-private partnerships for food redistribution has been analyzed in detail by Klitgaard (2011). In it, coordination between private technology companies and governments contributes to effectively managing surplus foods through collaboration. In the analysis, information platforms have been seen to act as a remedy for the loss of foods, with transparency and efficiency in tracking foods in surplus contributing towards increased effectiveness in distribution. Successful case studies, in addition, have exhibited that incentives for private intervention in food distribution networks make them strong. Implementation, however, is challenged by bureaucracies and misalignment of stakeholders. Technology-facilitated collaborations have contributed to fixing food security, and overcoming bureaucracies and strengthening partnerships could go a long way in strengthening distribution efforts. Existing studies prioritize information and communications technology (ICT) in minimizing consumption-related food loss and food waste. Adebisi (2023) considers ICT-led strategies for maximizing food redistribution, focusing on real-	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	40
	time tracking tools for surplus foods via digital technology. Smart inventories have minimized food loss, with AI-facilitated analysis maximizing distribution efficiency. In addition, donors' and takers' mobile platforms maximize efficiency with no wastage. Despite these, its difficulty comes with concerns regarding privacy and variable donors' willingness to use them. Observations have contributed to re-emphasizing overcoming technology-facilitated barriers and privacy in maximizing food redistribution. There is a future direction in studies towards developing trust and acceptance in technology-facilitated food networks. Technology interventions in the consumption production value chain contribute a lot to curbing food security, according to Grewal et al. (2024). In their view, proper use of technology tools can counteract loss and loss of food through proper predictive analysis and inventories. AI-powered demand forecasting is an effective tool in curbing excessive food, and proactive distribution planning for its distribution can counteract overall loss. Transparency through blockchain technology increases food traceability, and excess food reaches its target beneficiaries. However, High installation cost proves a barrier for small food distributors. Technology interventions with low cost can drive improvement in food distribution, and therefore, access to cheap technology tools can make them a part of more extensive food providers. One of the most significant trends in food redistribution is the heightened use of the Internet of Things (IoT) for efficient food waste management. Kumar and Prashar (2021) introduce a discussion regarding IoT-facilitated sensors and	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	41
	automation in improving food redistribution processes. In industries, real-time tracking is a key consideration in improving surplus food tracking, with intelligent sensors having been proven to maximize efficiency in logistics. Besides, cloud analysis maximizes decision-making in food redistribution networks. Notwithstanding that, cybersecurity vulnerabilities and concerns regarding data privacy restrict the widespread use of IoT technology. All these observations serve to re-emphasize security concerns in IoT-facilitated food redistribution. Augmentation of data protection could maximize trust and acceptance rates. Nandeesh et al. (2024) evaluate a range of technological innovations in food distribution and waste resolution. In terms of function, AI-matching platforms represent a key innovation in the efficient distribution of spare food. In its examination, distribution efficiency is maximized through automation, with a demonstration that algorithms in machines maximize networks for food donation. Besides, case studies confirm that integration with smart logistics reduces operational costs. Nevertheless, the digital exclusion of small providers curtails access to networks. Regarding these observations, inclusivity in digital tools maximizes gain in food distribution and can drive overall efficiency through access for small entities. Jagtap and Rahimifard (2019) have elaborated on IoT's contribution towards efficiency in food distribution channels with a specific view towards redistribution. In its article, smart logistics and its contribution towards minimizing food loss and efficiently using any spare food have been stressed. In its report,	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	42
	tracking through automation has been identified as a critical innovation, supporting that technological interventions streamline food logistics. Its conclusion supported such studies towards AI-facilitated distribution of spare food. Despite this, the high cost of installation and regulative restrictions serve as a deterrent to its widespread use. These observations validate that policy intervention can make IoT-based food redistribution even more efficient. Removal of regulative restrictions can make its use convenient and improve efficiency. Digital Transformation in Food Supply Chains Food safety aims to enable all people to have safe, nutritious, and enough food for their dietary needs and preferences for a healthy and active life (Zhou et al., 2021). Digital transformation has significantly reshaped food supply chains, addressing challenges like food waste, inefficiencies, and food insecurity, particularly in urban areas. Technological tools like smartphones that can be used daily, such as the Internet of Things (IoT), blockchain, artificial intelligence (AI), and big data analytics, have emerged as critical enablers of efficient and transparent food systems. Also, these innovations can potentially mitigate barriers in food distribution networks, especially in underserved urban settings like Mandaluyong City, by optimizing logistics and improving access to nutritious food. Blockchain technology has emerged as a cornerstone for improving transparency in food supply chains. Blockchain addresses critical concerns about food safety and consumer trust by allowing stakeholders to trace food origins and	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	43
	verify quality. Singh et al. (2022) say that one has many databases or stakeholders that could cause information asymmetry and possible loss of information during the transition. This has resulted in several cases related to false claims made by packed food companies, bringing economic loss and impacting consumer trust to a large extent. Blockchain's robust and decentralized functionality can be used to tackle food fraud and security. Capistrano (2021) highlights the application of blockchain in tracking food sources, enabling accountability and ensuring compliance with safety standards., the challenges faced in urban contexts, and their impact on food security. Advancements in IoT, blockchain, and AI have driven the digital transformation of food supply chains, collectively enhancing efficiency and reducing waste. According to Sharma and Singh (2022), blockchain technology is pivotal in ensuring transparency and building consumer trust by providing a tamper-proof record of food origins, processing, and distribution; IoT-enabled devices allow real-time monitoring of food conditions during transport, ensuring quality and reducing spoilage but also improve overall inventory management, creating a more efficient supply. Additionally, blockchain prevents food fraud and unethical practices by ensuring that perishable goods are stored and transported under proper conditions. Blockchain chatbot offers customers a transparent and safe platform for communication by fusing the conversational skills of chatbots with the power of blockchain technology, which will form mutual trust. Chatbots can securely	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	44
	record and authenticate transactions, verify user identities, and enable peer-to-peer communication without intermediaries by utilizing the blockchain's immutable ledger (Mechkaroska and Domazet, 2024). This is particularly relevant in urban settings where food safety concerns can deter consumers from fully embracing digital food delivery platforms. Similarly, Zhou et al. (2021) highlight AI's importance in improving supply chain decision-making processes. These AI systems can analyze vast datasets to predict demand, optimize routes, and identify inefficiencies in distribution networks. This integration of technology not only improves operational efficiency but also supports sustainability by minimizing food waste. Mechkaroska and Domazet (2024) further explore innovative approaches, proposing the integration of blockchain with IoT and chatbots to revolutionize food delivery systems. These tools enable communications between stakeholders, optimize delivery logistics, and provide real-time updates to consumers. Such advancements are particularly relevant in urban settings, where the demand for timely and reliable food delivery is high. Globally, food supply chains have long grappled with inefficiencies, including food waste, lack of transparency, and logistical challenges. The rise of digital technologies offers a transformative solution, aligning with the United Nations' Sustainable Development Goals (SDGs), particularly Goal 2, which aims to achieve zero hunger by 2030. Urban areas such as Mandaluyong City in the Philippines represent microcosms of the global food insecurity crisis. This will support Akinbamini et al. (2025) mention of	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	45
	the importance of supply chain collaboration in overcoming systemic inefficiencies. Their study in Nigeria demonstrates that effective communication and coordination among stakeholders are critical for leveraging digital tools. However, they also note that socioeconomic disparities and a lack of digital literacy often exacerbate inequalities within most food systems, underscoring the need for digital innovations. IoT gadgets and AI are extremely useful in the food delivery industry since they optimize delivery operations and guarantee that the quality and safety criteria are fulfilled. Barriers to Technological Adoption in Food Distribution Networks Digital transformation offers numerous consumer benefits, but several barriers hinder the widespread adoption of technological innovations in food distribution networks. These barriers include financial constraints, technological complexity, regulatory challenges, and resistance to change. Understanding these challenges is crucial in developing strategies to facilitate technology integration in the food industry. According to Akinbamini et al. (2025), one of the primary barriers to digital transformation in food supply chains is the high cost of technology adoption. Many small and medium-sized enterprises (SMEs) lack the financial resources to invest in IoT, blockchain, and AI-based solutions. This supports Talukder et al. (2022), who note that setting up a blockchain-enabled supply chain requires substantial investments in infrastructure, software, and personnel training making it inaccessible for smaller food distributors.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	46
	Furthermore, return on investment (ROI) concerns discourage some companies from fully digitizing their operations. Zhou et al. (2021) add that while digital transformation can lead to long-term efficiency gains, the initial costs are often prohibitive, particularly for businesses operating on tight profit margins, sometimes even forcing them to shut down operations in the short term. In addition, consumer preference for traditional marketing methods continues to dominate, posing another hurdle to digital adoption. A significant challenge is the resistance to change among stakeholders. Many traditional food supply chain operators prefer conventional methods and are hesitant to embrace digital solutions (Akinbamini et al., 2025). This reluctance often stems from a lack of digital literacy, fear of job displacement, and skepticism about technology reliability. Singh and Sharma (2022) suggest that educational workshops, subsidies, and regulatory support can help bridge the knowledge gap and foster a culture of innovation in food distribution networks. However, these interventions still require substantial investment, infrastructure, and training resources that small businesses typically lack. With the increasing reliance on digital platforms, food supply chains are also becoming more vulnerable to cyber threats. Kaur et al. (2022) emphasize that while blockchain enhances security, it is not immune to hacking, data breaches, and fraudulent transactions. The lack of uniform data protection laws across different regions further complicates compliance and discourages businesses	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	47
	from fully embracing digital transformation. As a result, companies must invest in robust cybersecurity measures, which only adds to the overall cost of digital integration. Regulatory and compliance challenges further exacerbate technological adoption issues. Governments impose strict food safety and data privacy regulations, requiring businesses to comply with complex legal frameworks. Wang et al. (2024) highlight that although blockchain-based solutions enhance transparency, they must still align with stringent legal mandates such as the General Data Protection Regulation (GDPR) and national food safety laws. Navigating these regulatory landscapes adds further complexity to digital adoption. These challenges are particularly pronounced in developing countries, where infrastructure limitations such as inadequate internet connectivity, unreliable power supply, and limited digital infrastructure hinder the implementation of digital solutions. Mechkaroska and Domazet (2024) emphasize the need for robust infrastructure, particularly in studies involving blockchain-integrated chatbots in food delivery. Amid these structural and systemic challenges, the increasing use of digital platforms for food access has notably transformed how households obtain and consume food, particularly through online ordering. According to Jabbour Al Maalouf et al. (2024), consumer decisions around food ordering are influenced by convenience, trust in the platform, and the perceived quality of food. The authors found that when digital platforms	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	48
	consistently deliver high-quality food, consumers are more likely to use them frequently. This link between perceived food quality and digital engagement is essential for understanding food security at the household level, particularly regarding food utilization and food stability. Food offered on digital platforms that is high in quality, safety, and nutritional value increases the credibility and acceptance of these platforms as viable household food sources. An important insight emerges from the urban orientation of the study: the perceived quality of food available through digital platforms directly influences food access decisions. This further underscores the importance of quality as a critical dimension within food security frameworks. Trust and E-commerce Acceptance in the Philippines Existing Philippine e-commerce studies have focused on the growing role of trust in shaping consumption behavior. Capistrano (2021) researched the fact that trust is a critical consideration in the acceptance and use of tourism books and food delivery platforms in the Philippines. Capistrano's contribution to food platforms proves that Philippine buyers are apprehensive about web transactions due to fraud, service failure, and lack of buyer protection. Vendor trust and perceived security of a platform have been proven in this work to act as critical factors in trust, with a demonstration that platforms with transparent policies and secure payment processes will have a high probability of eliciting repeat use. Besides, the role of review and a firm name in shaping trust in a platform are stressed in this work. Capistrano provides supporting evidence for trust	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	49
	developed through long-term service delivery, in agreement with studies arguing that repeat use is facilitated through positive past experiences. However, a weakness of this work is its failure to evaluate the role of government intervention in creating trust in a platform for buyers. These observations present a basis for trust concerns in Philippine electronic platforms, a dominant theme in this work. It underscores a necessity for security concerns and service dependability in enhancing trust in a buyer platform. The implications for this work lay in the necessity of additional evaluation of trust development in Philippine electronic platforms. By resolving such, electronic food platforms can contribute towards a high level of use and repeat use for a platform. A new development in studies in e-commerce is an increased concern for trust and its impact on electronic transactions. Belmonte et al. (2023) viewed trust is an important consideration in electronic platform acceptance in the Philippines, particularly in food delivery through electronic platforms. Belmonte et al. have extensively researched how payment platforms have been adopted in mobile apps and their impact on shaping trust in electronic payments. The current study confirmed that performance expectancy and ease of use significantly contribute to trust in digital food platforms, confirming that a platform will be utilized when friendly and free of glitches in use. Besides, social influence, social media, and peer recommendations considerably contribute to trust development, according to other studies. Additionally, they have an analysis confirming that trust in electronic	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	50
	commerce is directly proportional to security protocols embraced in electronic platforms. Capistrano's (2021) article confirms this, confirming that transparency and dependability significantly contribute to raising adoption levels. Nevertheless, Belmonte et al. confirm that apprehensions about fraud and scams in cyberspace have long been a strong deterrent to trust. All these studies contribute to explaining trust in a consumer and its impact on electronic commerce in the Philippines. This confirms a critical contribution towards overcoming security vulnerabilities and enhancing information dissemination for heightened trust in electronic food platforms. Online service providers imply trust development; for heightened use, they must have trust development programs in place. Current studies regarding trust in mobile commerce reveal the increased importance of security and platform trustworthiness. According to Teves (2023), trust in mobile buying apps is a key consideration in driving offline and online purchasing behavior in the Philippines. In food delivery platforms, a study conducted by Teves proves that buyers are not willing to use electronic platforms for fear of scams, security breaches, and poor service delivery. In its analysis, such concerns over privacy and lack of supervision significantly affect trust, proving that buyers prefer platforms with effective security protocols. In addition, such additional observations confirm that feedback and review significantly shape buyers' perceptions. Teves offers proof that secure payment gateways and third-party verification processes have a bearing on trust establishment. In agreement	

with Capistrano (2021), such trust is increased when platforms have transparent policies.

Nevertheless, Teves observes that despite such controls, buyers' lack of trust continues, particularly with older buyers. Such observations explain buyers' reluctance regarding mobile buying and electronic food platforms in the Philippines. Such observations confirm the increased importance of dealing with concerns over data privacy and compliance with regulators in enhancing trust. Observations in such studies confirm that increased security protocols and transparency could have a significant impact on enhancing electronic buying in general.

The application of e-commerce in Manila has been examined in detail to comprehend trust in electronic transactions. Consumer trust continues to serve as a driving force in the use of e-commerce, particularly in web and mobile platforms, according to Lim (2014). In the case of food delivery platforms, investigations conducted by Lim indicate buyers utilize experiences and word of mouth when choosing platforms for use. In its examination, platform trust and familiarity with a brand have been found to affect trust, confirming that buyers prefer dealing with companies with familiarity. Besides, additional findings confirm trust in return policies and money-back options in minimizing hesitation in purchasing, according to Lim. There is evidence documented by Lim that perceived risk constrains trust development, which agrees with additional studies stating that rogue seller concerns restrict use. Despite this, a disadvantage of

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	52
	such a study is that it cannot cover in detail the rapid development in security in mobile payments and legislation for protecting buyers. These studies serve as a basis for understanding trust in Philippine electronic trading platforms. It demonstrates the imperative for delivering customer satisfaction and trust establishment methodologies. The business implications of such a study confirm that companies must prioritize transparency and credibility in developing confidence in electronic platforms for buyers. Existing trust mechanism studies in food delivery apps illustrate growing prominence in trust translation between offline and online experiences. According to an argument by Mai and Nguyen (2024), trust in food delivery platforms is closely determined through offline experiences with an equivalent brand name. In a digital food service study, work performed by Mai and Nguyen attests that familiarity with a store's offline presence positively reflects confidence in its online equivalent. In its examination, a principal trust builder in trust development, a brand name, was determined, with success proven for long-standing food chains competing in digital acceptance. In addition, ancillary findings verify the role of ongoing mobile app experiences in creating trust in buyers. There is corroborating evidence in the work performed by Mai and Nguyen, with positive word-of-mouth and mobile usability proving to counteract trust concerns. In agreement with Su et al. (2022), who determined that ease of use in perception plays a significant role in terms of acceptance, a drawback in terms of current work is a lack of	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	53
	consideration for newer, stand-alone food ventures with no offline presence, with an overreliance placed in terms of digital platforms alone. Findings in this work form a basis for investigating trust in food delivery platforms and its relation to familiarity with a name in a brand name. In its implications, a growing role for new ventures investing in trust development is stressed when competing with long-standing brands. A significant trend in e-commerce studies is growing concern with trust in mobile apps, particularly food delivery platforms. Su et al. (2022) report that dependability, security, and service efficiency determine trust in mobile food delivery platforms. Su et al. have examined AI-powered recommendations in food platforms in detail to understand their contribution to consumer decision-making processes. In their investigation, Su et al. determined that personalized recommendations build trust in consumers, with the observation that users trust platforms that offer personalized recommendations more. In addition, other observations reveal that real-time tracking plays a significant role in raising trust levels on platforms. Su et al. present evidence that platform responsiveness is closely related to user satisfaction, supporting a similar observation by Wiencierz and Lünich (2022) that transparency in information use raises trust in platforms. Su et al., however, report that trust in platforms is a concern in delayed delivery and service variation cases. In conclusion, these observations form a basis for explaining the contribution of platform features to trust in food delivery platforms. As such, it	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	54
	underlines the necessity of platform service efficiency to build confidence in platforms for users. As a consequence, the implications of such a study reveal that platforms must correct service variation for trust to prevail in food delivery platforms. Current trends in open data studies confirm the growing prominence of transparency in creating trust with consumers. Wiencierz and Lünich (2022) argue that trust in platforms is predominantly determined through transparency in managing information for and about users. In food delivery platforms, a study conducted by Wiencierz and Lünich confirms that platforms with transparent information regarding the use of information are highly willing to use them, consumers. In discovering transparency in pricing and policies, trust is high, and through demonstration, it is confirmed that information disclosed to them is in a high level of security for consumers. Besides, additional observations confirm that platform accountability creates confidence in platform trust. As argued by Wiencierz and Lünich, compliance with laws and ethics in business creates a positive perception of trust in terms of trustfulness in platforms. According to Su et al. (2022), transparency in service reduces suspicions in platforms. Despite the drawback of this study not including review platforms in trust development, all these observations form a basis for creating an understanding of transparency in trust with consumers. This confirms using transparency and compliance with laws to create platform confidence. According to Su et al. (2022), implications in this study confirm that food delivery platforms need to use transparent policies to create trust with consumers.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	55
	The differentiation between electronic and traditional commerce has been examined in detail to comprehend trust variation between such platforms. According to Balgomera et al. (2022), trust in electronic platforms is less compared to traditional retail, citing a lack of face-to-face contact and perceived danger. In food delivery platforms, a study by Balgomera et al. confirms that trust in electronic platforms is affected a lot by service dependability and payment security in electronic platforms. In such a study, reluctance to buy arose from accuracy concerns and fraud, confirming platform dependability as a firm trust variable. Besides, additional studies confirm that secure payment options enhance users' confidence. According to Balgomera et al., platforms with numerous payment options and buying protection policies enhance buyers' trust. According to Mai and Nguyen (2024), trust in platforms is enhanced when service delivery is constant and uniform. However, according to Balgomera et al., trust concerns arise when platforms and fraudsters sometimes cause service delivery failures. All such studies serve as a basis for the comprehension of buyers' concerns about electronic trust in platforms. This proves significant in enhancing platform dependability and security to instill buyer confidence. The implications of such studies confirm that electronic platforms in food delivery must have stricter controls in terms of service delivery in a move to enhance trust and buyers' confidence in such platforms.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	56
	Ethical, Social Implications and Role of Data Analytics in Tackling Hunger Big data has reshaped how food is managed, spanning production to distribution and tackling pressing concerns such as hunger and food waste. As Sonka (2015) describes, big data represents a vast reservoir of valuable insights from historical and real-time data sources. While the food industry has seen remarkable advancements in efficiency, challenges persist. Wolfert et al. (2017) emphasize that the need for precise data collection and updating as global populations grow becomes even more critical. Integrating distributed ledger technologies presents an opportunity to enhance data accessibility and sharing across food systems, ensuring that all stakeholders benefit from improved efficiency and decision-making. Jacobs et al. (2021) further argue that as digital collaboration in the food sector expands, the ethical dimensions of data sharing, particularly trust and transparency, must remain at the forefront of discussions. The Role of Big Data in Addressing Hunger According to Marvin et al. (2017), big data analytics has become vital in food distribution and hunger alleviation, mainly predicting demand and reducing food wastage. Machine learning models can analyze consumption patterns and forecast shortages, enabling food banks and governments to distribute resources more efficiently. Wolfert et al. (2017) highlighted that Internet of Things (IoT) devices and sensors monitor food freshness, improving food safety and reducing unnecessary waste. Despite these advancements, they pointed out that	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	57
	challenges remain in ensuring that data-driven solutions are ethical, fair, and accessible to all stakeholders in the food system (Jacobs et al., 2021). Trust, Transparency, and Accountability in Data Analytics Using big data in the food industry raises significant ethical concerns regarding trust and transparency. Wiencierz and Lünich (2020) emphasized that trust is a fundamental aspect of open data sharing, as individuals and organizations must be confident that their data will not be misused. Studies indicate that when users perceive transparency in data applications, they are more likely to contribute their data to collective efforts (Bot et al., 2018; Evjemo et al., 2019). However, Jacobs et al. (2021) warned that the lack of clear accountability in AI-driven decision-making poses risks, particularly when automated systems allocate food resources. Ethical and Social Implications of Data-Driven Food Systems One of the primary ethical challenges in digital collaboration is the disparity in data access and control. Bronson and Knezevic (2016) argued that large corporations that dominate the food industry often benefit disproportionately from big data, while smaller stakeholders, including local farmers and food distributors, may struggle to access the same insights. Gevaert et al. (2021) also raised concerns about biases embedded in machine learning algorithms used for food distribution, which may lead to unintended discrimination against marginalized communities. The lack of diverse representation in dataset development further	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	58
	exacerbates this issue, necessitating policies to promote fairness in AI applications (Paullada et al., 2021). Distributed Ledger Technologies and Data Ethics The emergence of distributed ledger technologies (DLTs), including blockchain, has been suggested as a way to enhance transparency and accountability within food supply chains. Jacobs et al. (2021) explained that DLTs enable decentralized data sharing while maintaining data integrity, allowing stakeholders to track food sources, expiration dates, and logistics in real-time. However, Jacobs et al. (2021) pointed out challenges in ensuring equitable participation in such systems. Smaller organizations may lack the resources to integrate blockchain solutions, leading to an imbalance in data governance. Policy Recommendations and the Future of Ethical Data Collaboration Policy interventions must address ethical concerns to ensure that big data contributes equitably to hunger alleviation. Gevaert et al. (2021) recommended that regulatory frameworks prioritize transparency in algorithmic decision-making and mandate inclusive data collection practices to mitigate biases. Additionally, fostering collaboration between governments, private entities, and non-profits can help bridge the digital divide and ensure that data-driven solutions benefit all stakeholders in the food ecosystem (Marvin et al., 2017). Ethical AI guidelines should be developed specifically for food systems, ensuring accountability mechanisms are in place to prevent data misuse.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	59
	Big data and machine learning have immense potential to combat hunger and food waste, yet ethical and social implications must be addressed to ensure responsible implementation. Trust, transparency, and accountability are essential in fostering data-sharing initiatives that benefit the broader society rather than serving the interests of a few dominant entities. The food industry can harness data analytics for sustainable and equitable food distribution by integrating ethical considerations into technological advancements. Case Studies of Technology-Supported Food Distribution Systems Current studies in food distribution exhibit growing significance in leveraging blockchain technology to offer transparency and efficiency. According to an argument posed by Holmberg and Åquist (2018), blockchain technology is critical in offering increased traceability and reduced fraud in food value chains. In a case study, they present that frameworks backed with blockchain technology have enhanced information security and streamlined logistics. In a study, real-time tracking of food distribution, through an added level of traceability, reduces food loss and ensures accountability. Besides, according to Holmberg and Åquist (2018), smart contracts in a blockchain system make a transaction automatic, offering perfect food distribution. Despite this, high installation costs and technological expertise requirements form a significant barrier to its use. In conclusion, such studies exhibit growing significance in leveraging a secure and	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	60
	efficient food value chain through blockchain technology. Overcoming barriers to its use could significantly contribute to improving food distribution networks. Integration with digital nutrition informatics for access and efficiency is one key food distribution system trend. Ayres and Hoggie (2012) investigate the use of digital tools in regulating food distribution and nutritional value tracking. In its report according to them, real-time tracking of food distribution via digital platforms confirms that information-intensive strategies maximize the distribution of resources. Besides, in its report, AI's application in nutrition analysis in food distribution networks, with an eye towards automation, is seen. According to such studies, obstacles in terms of digital exclusion and information complexity inhibit its actualization. Such studies unveil the role of digital nutrition informatics in food redistribution. Eliminating digital access obstacles could make such a system practical. Public food distribution networks have been extensively researched for integration with blockchain technology by Kumar (2021). In his article, efficiency and transparency in food supply chains, particularly in crises, have been examined through a review of integration with blockchain technology. The research confirmed that public distribution networks with blockchain technology reduce fraud and inefficiencies and that secure electronic books promote accuracy in food distribution. Besides, intelligent integration, according to the research, aids in tracking foods in an automated form and reduces food distribution delays. Notwithstanding its efficiency, restrictions such as laws and	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	61
	infrastructure investments restrict its use significantly. All these confirm that food security is important and can be boosted through blockchain technology, and overcoming restrictions and infrastructure can maximize its use in food distribution networks. Haque and Kohda (2016) explore technology-facilitated food safety management in distribution networks. According to them, computer tools have to be adopted to enhance quality assurance and counteract foodborne danger. In its study, computerized checking contributed to enhancing compliance with food security, confirming that computerized checking reduces the danger of contamination. Real-time tracking of storing and transporting foods' conditions aids in compliance with requirements for security, in addition to computer checking. Despite its application, its difficulty in its widespread use arises from its resistivity towards computer transformation and its expense, acting as a barrier to its use. All these reveals why technology-facilitated food security management is imperative. Overcoming cost and acceptance barriers could make food distribution secure and efficient. Bhat and Dubey (2023) mention that food value chains have been re-engineered with efficiency gain and increased traceability through blockchain technology. In this case, consideration is taken for using blockchain in food distribution and security improvement in Walmart. The current studies support using blockchain platforms for traceability, reducing loss and fraud in foods, and real-time tracking maximizes accountability. Besides, Walmart's success in its	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	62
	use of food security validates its use at a more significant level. However, its high installation price and its demand for harmonized regulative frameworks inhibit its use at a more significant level. Therefore, all these factors make its efficiency in food distribution even more extensive in terms of viability. Removing cost and regulative obstacles could maximize its use in global supply chains. Barnett et al. (2022) have argued that technology-facilitated nutrition platforms and technology-facilitated platform effectiveness in food delivery in their work, technology-facilitated platforms for tracking and optimizing food delivery to high-risk groups have been prioritized, in their opinion. The review enables real-time analysis of information to maximize accuracy in food distribution, predictive modeling to maximize efficiency, and, in addition, electronic programs for health make nutritional requirements for foods delivered at scattered locations a reality. Integrating programs for health with food distribution in a complex manner, nevertheless, is challenging. The information below describes future food distribution through digital health technology. There is a future for studies that will make alternatives for the harmonious integration of food and health systems a reality. Kamilaris et al. (2019) explore the development of value chains in agriculture and food with an eye on developing food distribution networks. About them, platforms with a basis in blockchain make value chains transparent, with real-time tracking working towards minimizing inefficiencies. As a secondary observation, work identifies a role for decentralized ledgers in anti-food fraud	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	63
	activity. Despite that, interoperability between competing platforms, for example, creates a barrier to use. Observations work towards reenacting a role for protocols in a standardized form for developing food distribution networks. Overcoming interoperability could make a system efficient and scalable. Abeltino et al. (2025) review technology for tracking diets and food distribution. Their review discusses technology for optimized food distribution according to individual nutritional requirements in their article for precision nutrition. In its review and discovery, AI technology for tracking diets maximizes target food distribution, proving efficiency is boosted through data-intensive methodologies. Besides, mobile programs make nutritional guidelines and confirmative food aid accessible. Notwithstanding, poor communities' access continues to undermine its development. In conclusion, such a review proves a necessity for accessible technology for optimized precision nutrition in food distribution. Expanded access to such technology can maximize its contribution in terms of food security. Geospatial Technologies in Food Distribution This section discusses how Geospatial technologies, such as Geographic Information Systems (GIS), remote sensing, and spatial computing, have significantly enhanced food distribution networks by optimizing logistics, reducing food waste, and improving supply chain efficiency. The integration of these technologies has allowed stakeholders to make data-driven decisions, particularly in urban areas where food demand is high, and distribution efficiency	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	64
	is crucial (Ma et al., 2024). By leveraging spatial computing, food supply chains can become more responsive to disruptions, ensuring food reaches consumers more efficiently. One of the key applications of GIS in food distribution is its role in combating food insecurity. GIS technology enables real-time tracking of food supplies, helping organizations identify underserved areas and allocate resources effectively (Edinger, 2024). By mapping food deserts and understanding regional food availability, policymakers and humanitarian organizations can improve food accessibility, reducing disparities in food distribution. Additionally, geospatial technologies enable businesses and nonprofits to optimize their food distribution strategies. For instance, geospatial networking tools have been instrumental in identifying optimal locations for food sources, as demonstrated in studies focused on food availability in North Carolina (Mulrooney et al., 2021). Such tools enhance logistical planning, reduce transportation costs, and minimize delays in food delivery. Recent advancements have also explored the intersection of geospatial technologies and innovative food delivery platforms. On-demand food delivery services increasingly rely on micro-location analytics to optimize store placements, ensuring fast and efficient service (Hu et al., 2024). These developments underscore the potential of digital transformation in food logistics to enhance consumer convenience while improving operational efficiency. Despite these advancements, implementing geospatial technologies	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	65
	in food distribution presents challenges. High costs, data privacy concerns, and interoperability issues hinder widespread adoption, particularly for small-scale businesses and developing regions. Addressing these barriers requires further innovation, investment in geospatial workforce development, and supportive policies (Alqarni, 2020). As geospatial technologies evolve, their role in food distribution will likely expand, driving efficiency, sustainability, and resilience in the global food supply chain. Synthesis The reviewed literature consistently highlights the transformative impact of digital technologies on improving food access and addressing food insecurity. Across multiple studies, emerging technologies such as artificial intelligence (AI), Internet of Things (IoT), and mobile platforms are shown to enhance food distribution efficiency, particularly for vulnerable populations like low-income households, the elderly, and rural communities (Sharma et al., 2024; Hwang et al., 2024; Yaiprasert & Hidayanto, 2024). These innovations align with sustainability goals and present long-term solutions to systemic food access issues. However, several scholars caution that these benefits are not equally experienced due to persistent digital divides related to infrastructure, affordability, and digital literacy (Podkolzina et al., 2023; Vedovato et al., 2022). A shared concern among researchers is that technological solutions, while promising, are limited by user acceptance. Factors such as trust, transparency,	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	66
	and convenience significantly influence whether consumers adopt food technologies (Garrido & Gallardo, 2022; Kamboj et al., 2023; Teves, 2023). For example, while online grocery apps and delivery services improve the availability of nutritious foods (Fernandez & Raine, 2021), these services are often inaccessible to economically disadvantaged users due to digital hesitancy, cost, or lack of regulation (Singh & Sharma, 2022; Mechkaroska & Domazet, 2024). A consensus has emerged that technical functionality alone does not guarantee equitable usage, as socioeconomic barriers continue to be a central issue. Affordability, in particular, emerges as a key obstacle across the literature. Even with advanced logistics and digital tools, many low-income consumers struggle to access nutritious food if prices remain high (Chungchunlam & Moughan, 2024; Herforth et al., 2020; Masters et al., 2023). While some authors propose innovations such as AI forecasting and intelligent logistics to reduce costs (Igwe et al., 2024; National Academies, 2022), some argue that policymakers must complement these strategies with inclusive interventions to ensure their effectiveness. When it comes to technical solutions to reduce food waste, scholars point to blockchain, redistribution platforms, and geospatial mapping as efficient tools for reallocating surplus food (Olajide et al., 2024; Tian et al., 2025; Ma et al., 2024). However, others raise concerns that overreliance on technology can neglect critical issues such as privacy, data bias, and cultural fit (Gille, 2012; Granheim et al., 2022; Wu et al., 2022). These contradictions suggest that while	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	67
	innovation is necessary, ethical and localized considerations must guide development. In the Philippine context, studies have revealed that platform trust, brand familiarity, and peer recommendations strongly influence the adoption of digital food services (Capistrano, 2021; Belmonte et al., 2023; Mai & Nguyen, 2024). This insight complements global findings on trust but adds a cultural dimension that emphasizes the need for localized platform design and marketing strategies. Finally, scholars draw attention to the ethical implications of big data and AI in food systems. While data-driven approaches can improve food targeting and distribution (Marvin et al., 2017; Wolfert et al., 2017), they must be balanced with responsible data governance, especially in avoiding algorithmic bias and reinforcing inequity (Jacobs et al., 2021; Bronson & Knezevic, 2016).	

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

CHAPTER III

RESEARCH METHODOLOGY

This chapter displays the Research Method Used, Population Frame and Sampling Scheme, Description of Respondents, Instruments Used, Data Gathering Procedures, and Statistical Treatment of Data.

Research Method Used

The researchers employed a quantitative research design with a descriptive-correlational approach to assess household food security levels. The descriptive component systematically outlines patterns and analyzes statistical relationships regarding the operation of digital food distribution platforms in urban areas, particularly in Mandaluyong City. In contrast, the correlational aspect seeks to explore the relationship between digital platforms and household demographics and socioeconomic status, determining the relationships between key variables, such as technology accessibility, trust, and food security (Creswell & Creswell, 2018).

To collect information from groups of people, a survey research method was employed to gather primary data from consumers, ensuring the systematic collection of quantifiable data regarding respondents' experiences with digital food distribution platforms, their perceptions of food security, and potential barriers to adopting such technologies.

A quantitative research approach ensures the objectivity, reliability, and generalizability of the findings (Creswell & Creswell, 2018). This approach

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	69
	enables statistical tools to analyze data efficiently, resulting in a more structured and replicable study. By focusing on numerical data and structured questions, this study minimizes subjectivity and ensures high accuracy in interpreting results. Furthermore, quantitative research has been widely used in studies examining technology adoption and food distribution systems, reinforcing its applicability in this context. The approach establishes statistically significant conclusions about a population by studying a representative sample (Lowhorn, 2007). Price and Lovell (2018) emphasize the importance of careful planning in research to prevent misinterpretations and inaccurate reporting of findings. Misleading quantitative study findings can have serious repercussions, including the misappropriation of public funds for ineffective policies and the exposure of service consumers to hazardous or inadequate treatments. The descriptive-correlational design is instrumental in identifying patterns and trends within food distribution networks. By describing current practices and examining the relationships between various variables, this study offers valuable insights into how digital food platforms affect food security in urban settings. Given its effectiveness, this approach was deemed most suitable for achieving the study's objectives (Lau, 2017; Seeram, 2019). Population Frame and Sampling Scheme As of 2020 data, Mandaluyong City recorded 425,758 population within 27 barangays including: Addition Hills, Bagong Silang, Barangka Drive, Barangka Ibaba, Barangka Ilaya, Barangka Itaas, Buayang Bato, Burol, Daang Bakal,	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	70
	Hagdang Bato Itaas, Hagdang Bato Libis, Harapin Ang Bukas, Highway Hills, Hulo, Mabini J/Rizal, Malamig, Mauway, Namayan, New Zaniga, Old Zaniga, Pag-asa, Plainview, Pleasant Hills, Población, San Jose, Vergara, and Wack-Wack Green Hills (2020 Census of Population and Housing Results, City of Mandaluyong Philippine Statistics Authority National Capital Region, 2021). The researchers used Cochran's formula to actualize respondents and balance cost and precision in a large population in Mandaluyong City, so they decided to obtain 400 respondents, where their precise level is ± 1, the estimated proportion is .20, and the confidence level of 96%, making the understudy in the population quite accurate in the measurement for household food insecurity access scale and their independent variables which understudy. Furthermore, 400 total respondents is already enough total number of respondents for most research; this supports the old theorem about the law of large numbers by Bernoulli (1713), from standard ± 5 precise level, estimated proportion of .5 and confidence level 95% in Cochran's formula, the researchers made a decision to decrease the estimated proportion by 20% number of food platforms users and decreases the number to its precise level and increase the confidence level, which the results increase the samples locked into 400 total samples. They decided to increase the number of samples, so the study approaches its mean. On the other hand, the researchers employed a multistage sampling scheme, including purposive and stratified sampling, to select the population in	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	71
	Mandaluyong, specifically in Barangay Addition Hills, Barangka Itaas, and Ibaba. Purposive sampling can make the researchers' study more time-efficient and relevant, allowing them to gain insight through a survey regarding the usage of food applications. Additionally, it is a purposive sampling method because the researchers provided criteria for selecting participants rather than randomly selecting individuals who do not use food applications and those who do. Instead, the researchers chose to survey people who used different food applications to know their daily level of hardships in ensuring food stability. Furthermore, Stratified sampling identifies which participants are involved in the survey, narrowing the profiles in each barangay to include people of different ages, education, and other specific demographics and socioeconomic characteristics in this research. Regardless of age, the researchers conducted surveys randomly as long as the consumers could use the food applications. Description of the Respondents The respondents in this study are from Mandaluyong City, specifically from Barangay Addition Hills, Barangka Ibaba, Barangka Itaas, and Barangka Drive, who actively or frequently use food applications as their primary or secondary food access. These respondents are selected based on their demographics and socioeconomic factors, including age, gender, employment status, Monthly Household Income, and Occupation (if employed). In the demographic results for the 18 to 24 years old age group, researchers recorded a total of 49 respondents. The age group 25 to 34 had 123 respondents,	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	72
	while those aged 35 to 44 numbered 57. In the 45 to 54 age group, there were 36 respondents, and 35 individuals aged 55 and above participated. Regarding gender, there are 213 males, 164 females, and 24 respondents who chose "prefer not to say." These demographics help them understand how applicable and relevant food platforms are to different groups. The socioeconomic results for the income bracket groups indicate the following ranges and corresponding numbers: below ₱10,000 (173 respondents), ₱10,001-₱20,000 (119 respondents), ₱20,001-₱30,000 (64 respondents), ₱30,001-₱40,000 (23 respondents), ₱40,001-₱50,000 (6 respondents), and above ₱50,000 (15 respondents). In the employment category, researchers grouped respondents as follows: 159 identified as Full-Time, 51 as Part-Time, 65 as Self-Employed, 56 as Unemployed, and 69 as Students. When asked about their occupations, 67 respondents chose the Service Industry, 58 identified as Office/Corporate, 38 as Entrepreneur/Business Owner, 61 as Manual Labor/Skilled Worker, and 20 marked Other. Additionally, 156 responses indicated that they do not confide, representing over 38% of the population. This survey aims to assess the socioeconomic characteristics of income, employment categories, and occupations to understand how respondents use food platforms, their motivations, and the specific occasions or circumstances that lead them to seek convenience through these platforms.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	73
	Instruments Used The research team designed the questionnaire for this study. It was validated by experts from the Departments of Information Technology, Food Technology, and Political Science, as well as a professional well-being coordinator before deployment. They used the questionnaire as the primary data collection instrument to assess the role of digital technology in food distribution networks. This survey aimed to gather demographic information, access to technology, and trust in digital food platforms. The team followed a transparent development process for this instrument. 1. Demographic and Socioeconomic This Section Collects participant details such as age, sex, employment status, monthly household income, occupation, and frequency of online food delivery use. These variables help segment respondents into relevant categories, aligning with their study's stratified and cluster sampling approach. 2. Household Food Insecurity Access Scale (HFIAS) This Section evaluates respondents' food insecurity, it seeks to identify the population of food-insecure people, and assesses their impact on the studies. It asked how often do the respondents experience food insecurity with the scale and ranges assessed.	

Scale	Ranges	Verbal Interpretation
1	1.00-1.74	Never
2	1.75-2.49	Rarely
3	2.50-3.24	Sometimes
4	3.25-4.00	Often

In Section 3: Factors Influencing Online Food Delivery Usage and Section

4: Barriers to Online Food Delivery Adoption, researchers measure the extent of agreement on a scale ranging from Strongly Agree to Disagree Strongly.

Scale	Ranges	Verbal Interpretation
1	1.00-1.74	Strongly Disagree
2	1.75-2.49	Disagree
3	2.50-3.24	Agree
4	3.25-4.00	Strongly Agree

3. Factors Influencing Online Food Delivery Usage

This section emphasizes the perceived ease of use and usefulness of technology to consumers, identifying the importance of convenience and accessibility, affordability and cost considerations, digital literacy, payment preferences, as well as food quality and availability.

4. Barriers to Online Food Delivery Adoption

This section highlights barriers that deter consumers from using digital food platforms and explains why consumers are indifferent towards them.

Reliability testing is conducted using Cronbach's alpha, a Statistical measure that assesses the consistency of Likert-scale items. Cronbach's alpha

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	75
	value of 0.70 or higher was considered acceptable, indicating that the survey items accurately captured the respondents' perceptions. This validation process ensures that the data collected is credible, with no errors, or at least a 95% confidence level (0.05 margin of error), and that research findings are improved. Data Gathering Procedures The data-gathering procedure involved three phases for surveying 400 respondents: Preliminary Phase. Academic advisors and an ethics review board approved the study, ensuring compliance with research ethics and data privacy regulations. A pilot survey of 35 respondents was conducted to test the reliability and validity of the questionnaire, and the questionnaire was refined based on the feedback received from respondents. Data Collection Phase. To reach a diverse sample, surveys were distributed online and in print. Online surveys were disseminated through digital platforms, ensuring accessibility for tech-savvy respondents. Respondents were informed about the study's objectives, ensuring voluntary participation with informed consent and maintaining confidentiality and anonymity. Data Processing and Analysis. Collected responses were encoded into statistical software (SPSS v.28) for analysis. Data cleaning was performed to ensure accuracy, removing incomplete or inconsistent responses. The final	

dataset was subjected to statistical analysis to draw meaningful conclusions regarding digital food platform adoption and its impact on food security.

Statistical Treatment of Data

Researchers used the following statistical tests to analyze and interpret the data gathered in the survey:

1. Frequencies (Manikandan, 2011) – A table that shows the distribution of class categories and observations divided by the range (from maximum to minimum) called "class interval."

2. Percentages (Cole & Altman, 2017) – Calculating the samples and mean to express the difference of the fraction as a whole.

$$\text{Percentage} = \left(\frac{\text{Whole}}{\text{Part}} \right) \times 10$$

Where:

Part = the portion or amount of comparison

Whole = the total or overall amount

x100 = converts the result into a percentage

The Statistical Treatment of Data used has been displayed in Tables 1 to 6, specifically, the distribution of respondents by Age, Sex, Employment Status, Income Level, Occupation, and Frequency of Online Food Delivery Consumption using Frequencies and Percentages.

3. Mean (Drummond & Tom, 2011) - the sum of all values in a dataset divided by the total number of values, or known as the average.

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

$\sum X$ is the sum of all values

N is the total number of values

4. Standard Deviation (Omda & Sergent, 2024) – A measurement of a data set's spread, from which it derives whether it is from the entire population or the sample population.

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Where:

σ = Standard Deviation

$\sum$ = Summation Symbol

x_i = Each data point in the sample

$\bar{x}$ = Sample means

$(x_i - \bar{x})^2$ = Squared deviation

$x_i - \bar{x}$ = Deviation of a data point from the mean

n = Sample size

$n - 1$ = Degrees of freedom

Researchers computed the Mean and Standard Deviation for variables related to Convenience and Accessibility, Affordability and Cost Considerations,

Digital Literacy and Payment Preferences, Food Quality and Availability, and Barriers (see Tables 7 to 15). This analysis describes the average responses and highlights the variability in perceptions across different households.

5. T-test (Bevans, 2023) - t-test is a parametric test used to compare the means of two groups to determine if they differ significantly from each other. It can also be used in hypothesis testing with the null hypothesis and alternative hypothesis (where the difference between the group means is zero or the difference is not equal to zero).

A t-test was used to determine if there is a significant difference in factors influencing respondents' use of online food delivery when grouped by sex. To answer the Statement of the Problem No. 6 and Statement of the Problem No. 7.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Where:

$\bar{x}_1, \bar{x}_2$ = sample means of group 1 and 2

s^2 = pooled standard error of the two groups

n_1, n_2 = number of observations in each group

6. ANOVA (Kenton, 2025) - ANOVA is a statistical method used to determine whether there are statistically significant differences between three or more independent groups. This works by analyzing the variance within groups and comparing whether observed differences in sample means are actual differences or random fluctuations between groups.

ANOVA was applied to determine whether there are significant differences in respondents' perceived barriers to the adoption of online food delivery platforms when respondents are grouped according to demographic profiles, such as age, sex, household income, and occupation. Making it an appropriate statistical test for testing significant variations among three or more groups to answer Statement of the Problem No. 6 and Statement of the Problem No. 7.

$$F = \frac{MST}{MSE}$$

Where:

F = ANOVA coefficient

MST = Mean sum of squares due to treatment

MSE = Mean sum of squares due to error

CHAPTER IV**PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA**

In this chapter, the data is presented and analyzed in tabular format. The results from the statistical calculations were interpreted using quantitative analysis and organized systematically in relation to the Statements of the Problem.

**Table 1 Frequency and Percentage Distribution of the Respondents
According to Age**

Age	Frequency	Percentage
18-24	149	37.25
25-34	123	30.75
35-44	57	14.25
45-54	36	9
55 and above	35	8.75
Total	400	100

Table 1 shows the frequency and percentage of respondents based on their age. the data show that the majority of respondents are between the ages of 18 and 24, with a total of 149 (37.25%), while the lowest age group is 55 and above, with a total of 35 (8.75%). This suggest that younger individuals were more represented in the survey compared to older age groups. This may indicate a greater engagement or accessibility of younger participants to the survey platform.

Table 2 Frequency and Percentage Distribution of the Respondents**According to Sex**

Sex	Frequency	Percentage
Male	213	53.25
Female	164	40.75
Prefer not to say	24	6
Total	400	100

As seen in Table 2, males with 213 (53.25%) are the population's major respondents in the four chosen barangays: Barangka Ibaba, Barangka Itaas, Barangka Drive, and Addition Hills. Female acquired only 164 or 40.75% of the population, whereas the rest preferred not to say their sex. This indicates that males were more represented among the respondents in the surveyed areas.

Table 3 Frequency and Percentage Distribution of the Respondents**According to Employment Status**

Employment Status	Frequency	Percentage
Full-Time	159	39.75
Part-Time	51	12.75
Self-Employed	65	16.25
Unemployed	56	14
Students	69	17.25
Total	400	100

As reported in Table 3, Employment Status, which describes whether the population is a Full-Time employee, Part-Time, Self-Employed, Unemployed, or Student, the highest number of respondents is mainly Full-Time employment, with

159 (39.75%), and the lowest number is partly employees, which is Part-Employee, obtained 51 or 12.75%.

Table 4 Frequency and Percentage Distribution of the Respondents

According to Income Level

Monthly Household Income	Frequency	Percentage
Below 10k	173	43.25
10,001-20,000	119	29.75
20,001-30,000	64	16
30,001-40,000	23	5.75
40,001-50,000	6	1.5
Above 50,000	15	3.75
Total	400	100

The Frequency and Percentage of the Respondents' Income Level table shows that an income of below ₱10,000 is the most common, with a total of 173 respondents, or 43.25%, and the least common income is between ₱40,001 and ₱50,000. However, the data also show that from the least common income, as the second-lowest uncommon household income, households can reach above ₱50,000. This indicates a significant variation of in income levels among the respondents. Although few in number, the higher – income households reflect a segment with stronger financial resources. Such income differences may contribute to unequal access to food, digital platform and basic services.

**Table 5 Frequency and Percentage Distribution of the Respondents
According to Occupation**

Occupation (If employed)	Frequency	Percentage
Service Industry	67	17
Office/Corporate	58	15
Entrepreneur/Business Owner	38	10
Manual Labor/Skilled worker	61	15
Others	20	5
Not Applicable (prefer not to say)	156	38
Total	400	100

Table 5 shows respondents occupation. The largest group was “Not Applicable” (n = 156, 38%). Among employed respondents, most were in the service industry (n = 67, 17%), followed by manual/skilled labor (n = 61, 15%) and office jobs (n = 58, 15%). The Fewest were business owners (n = 38, 10%) and others (n = 20, 5%). This distribution shows most respondents work in service or labor roles, with few involved in entrepreneurship.

Table 6 Respondent’s Frequency of Online Food Delivery Use

Online Food Delivery	Frequency	Percentage
Daily	35	8.75
Weekly	128	32
Monthly	59	14.75
Rarely	146	36.5
Never	32	8
Total	400	100

The data on online food delivery services shows some intriguing trends. Approximately 36.5% of respondents say they rarely use these services, while

	RIZAL TECHNOLOGICAL UNIVERSITY																			
	COLLEGE OF ARTS AND SCIENCES	84																		
	32% use them every week. A smaller portion, 14.75%, places orders monthly, and just 8.75% use online delivery daily. Interestingly, 8% of respondents report that they never use these services. Despite the availability of online food delivery options, many people do not use them regularly. Many individuals consider online food delivery convenient, but they typically use it only occasionally. They prefer home-cooked meals or shopping at local stores instead. This observation supports the conclusion by Yaiprasert and Hidayanto (2024), which highlights that, although online food platforms offer personalization and ease of use, various barriers limit their overall adoption. In addition, cultural habits and routines around food preparation may also influence user preferences. Factors such as concerns about trust, fraud, and platform security—highlighted by Capistrano (2021)—play a significant role in the infrequent use of these services. Moreover, some users express dissatisfaction with delivery cost and food quality, further discouraging regular use. Table 7 Respondent’s Experience of Food Insecurity <table> <tr> <th>Level of Food Insecurity</th><th>Frequency</th><th>Percentage</th></tr> <tr> <td>Never</td><td>162</td><td>41</td></tr> <tr> <td>Rarely</td><td>188</td><td>47</td></tr> <tr> <td>Sometimes</td><td>44</td><td>11</td></tr> <tr> <td>Often</td><td>6</td><td>1</td></tr> <tr> <td>Total</td><td>400</td><td>100</td></tr> </table>	Level of Food Insecurity	Frequency	Percentage	Never	162	41	Rarely	188	47	Sometimes	44	11	Often	6	1	Total	400	100	
Level of Food Insecurity	Frequency	Percentage																		
Never	162	41																		
Rarely	188	47																		
Sometimes	44	11																		
Often	6	1																		
Total	400	100																		

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	85
	According to the survey results, 47% of respondents reported rarely experiencing food insecurity, while 41% stated they had never faced it. On the other hand, 11% indicated that they sometimes experienced food insecurity, and just 1% claimed to confront it often. These findings suggest that a majority of respondents have adequate access to food, indicating that food insecurity is not a significant challenge for most. This level of food security may be due to stable incomes, government assistance, or other means that provide a consistent food supply, allowing nearly all individuals to meet their food needs effectively. However, it is essential to address the small percentage of people who experience food insecurity when planning support services. Research by Sharma et al. (2024) has analyzed stable food systems and their effectiveness in combating hunger. Additionally, Vedovato et al. (2022) found that low-income households with access to government or community aid are more likely to achieve food security. These studies illustrate that well-managed resource availability plays a crucial role in reducing food insecurity, as reflected in the survey findings. Such programs can prove valuable primarily in communities that are otherwise underserved. If there were to be mechanisms established for public – private collaboration, in the scaling – up and sustainability of these interventions would be eased even further. It is vital to continue investing food distribution infrastructure, education, and collaborations with communities. Moreover, keeping a consistent watch and collecting data o track assistance would further guarantee targeting of need.	

Table 8 Factors Influencing Household Use of Online Food Delivery in terms of Convenience and Accessibility

Convenience & Accessibility	Mean	SD	VI
1. Online food delivery makes it easier to access meals when I am too busy to cook.	1.71	0.65	Strongly Disagree
2. I prefer ordering food online because it saves me time.	2.02	0.708	Disagree
3. The availability of 24/7 food delivery services influences my decision to use these platforms.	1.88	0.739	Disagree
4. Online food delivery helps me access food options that are not available nearby.	1.84	0.681	Disagree
5. I find food delivery more convenient than grocery shopping or cooking at home.	2.03	0.81	Disagree
Overall	1.89	0.528	Disagree

The recorded average reactions indicate a general disagreement about the convenience and accessibility of online food delivery services. Respondents do not strongly agree that meal delivery services are helpful during busy times, available at all hours, or easier than cooking or shopping, with an overall mean rating ($\bar{x} = 1.89$, $SD = 0.528$). The statement about delivery assistance during hectic periods received the lowest mean score ($\bar{x} = 1.71$, $SD = 0.65$), with several participants strongly disagreeing. The rest of the items also yielded low mean scores: saving time ($\bar{x} = 2.02$, $SD = 0.708$), availability of 24/7 service ($\bar{x} = 1.88$, $SD = 0.739$), access to food not nearby ($\bar{x} = 1.84$, $SD = 0.681$), and convenience over grocery shopping or cooking at home ($\bar{x} = 2.03$, $SD = 0.81$). Overall, people seem to view online ordering and food delivery as neither convenient nor

accessible, suggesting that these services cannot be relied upon for regular meals. This feedback highlights a need for improvement in areas such as expanding delivery options, lowering prices, and offering a more diverse menu to enhance user convenience. These findings support Teves (2023), who noted that many users avoid food applications due to concerns about scams and poor customer service. Hwang et al. (2024) highlighted that older users often face usability challenges due to the design of food delivery apps, particularly those with limited digital skills. Capistrano (2021) also supports this notion, stating that a lack of trust in a system significantly hinders its continued use.

Table 9 Factors Influencing Household Use of Online Food Delivery Services in terms of Affordability and Cost Considerations

Affordability & Cost Considerations	Mean	SD	VI
1.I consider online food delivery to be an affordable option for meals.	1.96	0.724	Disagree
2. Discounts and promotions encourage me to order food online more often.	1.91	0.757	Disagree
3. The delivery fee significantly affects my decision to use food delivery services.	1.92	0.712	Disagree
4. I am willing to pay extra for the convenience of food delivery.	2.08	0.781	Disagree
5. Ordering food online is often more expensive than preparing meals at home.	1.84	0.832	Disagree
Overall	1.94	0.485	Disagree

The findings gathered from the tables on Affordability and Cost for Online Food Delivery reveal a consensus among respondents that online food delivery

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	88
	is not a cost-effective choice for meals, as reflected in a mean score ($\bar{x}$ = 1.96, SD = 0.724, VI = Disagree). Additionally, they showed disappointment with the notion that discounts and promotions would lead to increased orders ($\bar{x}$ = 1.91, SD = 0.757), that delivery fees affect their decision-making ($\bar{x}$ = 1.92, SD = 0.712), and that they are willing to pay more for delivery ($\bar{x}$ = 2.08, SD = 0.781), as well as with the belief that ordering food online is more expensive than cooking at home ($\bar{x}$ = 1.84, SD = 0.832). Overall, the average ($\bar{x}$ = 1.94, SD = 0.485) suggests that affordability and cost considerations do not significantly motivate the use of online food delivery services among this sample. The relatively low standard deviation (0.485) indicates consistency in the participants' responses, meaning that most respondents shared similar views on the cost-related aspects of food delivery. The close clustering of values supports the reliability of the overall conclusion that these services are not perceived as affordable. Garrido and Gallardo (2022) emphasize that while convenience strongly drives technology adoption, it falls short if users have concerns about quality, health, and trust. Wu et al. (2022) note that in Asia, convenience and promotions also influence adoption, but cultural and economic factors play a crucial role as well. Chawla and Puri (2022) highlight that perceptions of safety and trust heavily outweigh cost considerations. Towse and Mauskopf (2018) identify affordability as a primary barrier to adopting new food technologies, while Chungchunlam and	

	RIZAL TECHNOLOGICAL UNIVERSITY																													
	COLLEGE OF ARTS AND SCIENCES	89																												
	Moughan (2024) pinpoint income and cost as significant obstacles to maintaining healthy diets. Table 10 Factors Influencing Household Use of Online Food Delivery Services in terms of Digital Literacy and Payment Preferences <table><tr><th>Digital Literacy & Payment Preferences</th><th>Mean</th><th>SD</th><th>VI</th></tr><tr><td>1. I find it easy to navigate food delivery apps.</td><td>1.86</td><td>0.655</td><td>Disagree</td></tr><tr><td>2. I prefer using cashless payment methods (e-wallets, credit/debit cards) for food delivery.</td><td>2</td><td>0.779</td><td>Disagree</td></tr><tr><td>3. My familiarity with technology influences how often I use food delivery apps.</td><td>1.93</td><td>0.706</td><td>Disagree</td></tr><tr><td>4. I would use online food delivery more if the apps were easier to use.</td><td>1.9</td><td>0.715</td><td>Disagree</td></tr><tr><td>5. I rely on reviews to assess food quality.</td><td>1.8</td><td>0.723</td><td>Disagree</td></tr><tr><td>Overall</td><td>1.9</td><td>0.518</td><td>Disagree</td></tr></table> The data reveals that respondents generally disagreed with statements regarding digital literacy and payment preferences in the context of food delivery apps. Most participants found navigating these applications challenging, as reflected in a mean score ($\bar{x}$ = 1.86, SD = 0.655, VI = Disagreed). Additionally, they showed a slight preference for cashless payment methods, such as e-wallets and credit or debit cards, with a mean score ($\bar{x}$ = 2.00, SD = 0.779). This indicates a need for simpler app interfaces and better digital education.	Digital Literacy & Payment Preferences	Mean	SD	VI	1. I find it easy to navigate food delivery apps.	1.86	0.655	Disagree	2. I prefer using cashless payment methods (e-wallets, credit/debit cards) for food delivery.	2	0.779	Disagree	3. My familiarity with technology influences how often I use food delivery apps.	1.93	0.706	Disagree	4. I would use online food delivery more if the apps were easier to use.	1.9	0.715	Disagree	5. I rely on reviews to assess food quality.	1.8	0.723	Disagree	Overall	1.9	0.518	Disagree	
Digital Literacy & Payment Preferences	Mean	SD	VI																											
1. I find it easy to navigate food delivery apps.	1.86	0.655	Disagree																											
2. I prefer using cashless payment methods (e-wallets, credit/debit cards) for food delivery.	2	0.779	Disagree																											
3. My familiarity with technology influences how often I use food delivery apps.	1.93	0.706	Disagree																											
4. I would use online food delivery more if the apps were easier to use.	1.9	0.715	Disagree																											
5. I rely on reviews to assess food quality.	1.8	0.723	Disagree																											
Overall	1.9	0.518	Disagree																											

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	90
	Respondents also felt that their familiarity with technology did not significantly impact their frequency of app use, indicated by a mean score ($\bar{x}$ = 1.93, SD = 0.706). They disagreed with the idea that they would use food delivery services more often if the apps were easier to use, scoring ($\bar{x}$ = 1.90, SD = 0.715). Furthermore, they expressed low reliance on reviews for assessing food quality, resulting in a mean score ($\bar{x}$ = 1.80, SD = 0.723). This suggest that even with easier interfaces or improved digital support, adoption may remain limited due to deeper issues like trust and food safety concerns Hence, the researchers also inferred a logical explanation for why the results in Table 6: Frequency and Percentage Distribution of the Respondents According to Online Food Delivery Consumption and highlighted by Capistrano (2021) and Hwang et Al. (2024) shows that they rarely use online food delivery services. The findings also simply technological improvements alone may not address the hesitation toward app usage. Others factors, such as cultural habits and past negative experiences, could play a role shaping user preferences. Overall, the average mean score across all items was ($\bar{x}$ = 1.90, SD = 0.518), highlighting a general disagreement with the notion that digital literacy and payment preferences influence food delivery app usage. Additionally, respondents demonstrated low trust in the quality and safety of food. This concern aligns with existing literature on food safety issues (Zhou et al., 2021) and the need for transparent food supply chains that incorporate technologies such as blockchain, AI, and IoT (Mechkaroska & Domazet, 2024).	

Table 11 Factors Influencing Household Use of Online Food Delivery**Services in terms of Food Quality and Availability**

Food Quality & Availability	Mean	SD	VI
1. The variety of food options available through delivery apps influences my usage.	1.85	0.643	Disagree
2. I trust the food quality of restaurants available on delivery platforms.	1.95	0.679	Disagree
3. I choose food delivery for the variety it offers.	1.92	0.694	Disagree
4. I avoid online food delivery if I am uncertain about food hygiene.	1.84	0.735	Disagree
5. I order food online mostly from restaurants I have tried before.	1.81	0.71	Disagree
Overall	1.87	0.483	Disagree

With $\bar{x} = 1.95$ (SD = 0.679, VI = Disagree) as the highest total mean, it indicates that most people do not trust the food quality on restaurants' online food delivery platforms, primarily due to various factors. It was mentioned by Jabbour Al Maalouf et al. (2024) that perceived food quality is based on the overall assessment of factors, such as safety, freshness, taste, presentation, nutritional value, and health standards, by the general public. Hence, these factors also influence people's judgment: they do not just perceive food quality. Thus, according to the consistency of answers from the data, with the lowest ($\bar{x} = 1.81$, SD = 0.710, VI = Disagree), most of them do not just go to a restaurant they are familiar with, but also try another as long as they perceive it as aligned with their assessments.

Table 12 Common Barriers to the Adoption of Online Food Delivery**Services in terms of Financial Barriers**

Financial Barriers	Mean	SD	VI
1. Online food delivery is too expensive for my budget.	1.97	0.69	Disagree
2. High delivery fees discourage me from ordering food online.	1.89	0.73	Disagree
3. The minimum order requirement in food delivery apps is a barrier for me.	2.11	0.86	Disagree
4. I avoid online food delivery services due to hidden charges (such as service fees and taxes).	1.91	0.78	Disagree
5. I would use online food delivery more frequently if it were more affordable.	1.89	0.76	Disagree
Overall Mean	2.01	0.49	Disagree

The show data indicates that respondents generally disagree that financial barriers significantly limit their use of online food delivery services. They do not consider online delivery too costly ($\bar{x} = 1.97$, $SD = 0.69$) or feel discouraged by high delivery fees ($\bar{x} = 1.89$, $SD = 0.73$), hidden charges ($\bar{x} = 1.91$, $SD = 0.78$), or unaffordability ($\bar{x} = 1.89$, $SD = 0.76$). The statement "The minimum order requirement in food delivery apps is a barrier for me" raised slightly more concern with an average score of ($\bar{x} = 2.11$, $SD = 0.86$), but people still generally disagreed. This reaction likely stems from the fact that most individuals are not using online food delivery systems, regardless of their financial capacity to place orders.

Consequently, the average score of ($\bar{x} = 2.01$, $SD = 0.49$) indicates that people do not consider financial concerns to be significant obstacles.

Akinbamini et al. (2025) highlight that a significant challenge for digital transformation in food supply chains is the cost of adopting new technologies. Many small and medium-sized enterprises struggle to secure funding for investing in advanced technologies, such as the Internet of Things (IoT), blockchain, and artificial intelligence (AI). Talukder et al. (2022) emphasize that implementing blockchain-enabled supply chains requires substantial investments in infrastructure, software, and personnel training, which makes these innovations inaccessible to smaller food distributors.

Table 13 Common Barriers to the Adoption of Online Food Delivery

Services in terms of Digital Barriers and Technical Issues

Digital Barriers & Technical Issues	Mean	SD	VI
1. I find food delivery apps difficult to use due to lack of digital skills or complexity.	2.27	0.80	Disagree
2. My lack of access to a smartphone or stable internet prevents me from ordering food online.	2.31	0.81	Disagree
3. I sometimes face payment issues when ordering food online.	2.28	0.81	Disagree
4. The lack of customer support discourages me from using food delivery platforms.	2.23	0.77	Disagree
Overall	2.27	0.63	Disagree

The table indicates that the statement, *"My lack of access to a smartphone or stable internet prevents me from ordering food online,"* received the highest

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	94
	mean score ($\bar{x}$ = 2.31, SD = 0.81, VI = Disagree), suggesting that respondents generally disagree with this notion. This interpretation means that not having a smartphone does not stop users from ordering food, as they have other ways to access these services. Furthermore, feedback from consumers indicates that inadequate customer support does not deter them from using online food platforms ($\bar{x}$ = 2.23, SD = 0.77), as they can still meet their food needs without relying heavily on these services. These findings underscore the digital divide in utilizing food platforms. According to Wu et al. (2022), older users and those with limited exposure to technology face challenges when using food delivery apps. The complexity of these apps and concerns about service reliability further deter less tech-savvy individuals, as noted by Chawla and Puri (2022). Akinbamini et al. (2025) highlight that inadequate infrastructure, such as poor internet connectivity and a lack of mobile devices, can significantly hinder the adoption of digital food systems. This issue is especially relevant in lower-income metropolitan areas, where access to reliable devices and networks is crucial for digital engagement. Singh and Sharma (2022) also note that efforts to digitize food supply chains face obstacles due to traditional customers' low digital literacy and reluctance to adopt change. Collectively, these studies support the information in Table 13, indicating that technological barriers, digital limitations, and financial constraints significantly impact access to online meal services. Continues user feedback should also guide the improvement of digital food service systems.	

Addressing these challenges requires collaboration among developers, policymakers, and community organizations. To enhance food security through digital platforms, focus on better structure, digital literacy, and user-friendly app design for diverse users.

Table 14 Common Barriers to the Adoption of Online Food Delivery

Services in terms of Service and Reliability Concerns

Service & Reliability Concerns	Mean	SD	VI
1. I have had negative experiences with late deliveries.	2.15	1.25	Disagree
2. I avoid food delivery due to past issues with incorrect or missing orders.	2.16	0.81	Disagree
3. I don't trust food delivery platforms to handle my food safely.	2.24	0.82	Disagree
4. Limited food choices in food delivery apps prevent me from using them.	2.15	0.75	Disagree
5. I avoid online food delivery because of concerns about food freshness.	2.1	0.81	Disagree
Overall	2.16	0.64	Disagree

The table indicates that the third statement has the highest mean ($\bar{x} = 2.24$, $SD = 0.82$, $VI = \text{Disagree}$), suggesting that users maintain a positive or neutral view of the service and reliability of food delivery platforms. Many respondents believe these platforms can safely handle food during transit. The selection of food choices available on delivery apps does not hinder users' access to meals, as they provide varied options ($\bar{x} = 2.15$, $SD = 0.75$, $VI = \text{Disagree}$). Additionally, negative experiences with late deliveries are not a significant concern for most users ($\bar{x} = 2.15$, $SD = 1.25$, $VI = \text{Disagree}$); they either have no past issues or

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	96
	have not allowed past problems to deter them from using these applications ($\bar{x}$ = 2.16, SD = 0.81, VI = Disagree). Similarly, concerns about food quality ($\bar{x}$ = 2.1, SD = 0.81, VI = Disagree) do not stop users from making purchases, likely depending on the magnitude of their reasons. Despite a generally favorable impression of these food applications among those surveyed, this does not guarantee that the entire population shares the same experience, especially given the imperfect nature of food delivery platforms. Researchers recognize that previous studies, including those by Su et al. (2022), have highlighted how inconsistent service quality and delayed deliveries can undermine customer confidence in food delivery services. The relevance of this finding continues today, as many users still view these platforms as unreliable. Although most consumers generally have a positive attitude towards food delivery services, their trust tends to wane after receiving inconsistent service or encountering issues. This decline in trust can discourage them from using these platforms in the future. This situation is particularly concerning for households experiencing food insecurity, as they depend on timely and accurate deliveries for their daily meals. Moreover, Balgomera et al. (2022) emphasize that service reliability and payment security are critical factors influencing user trust in digital platforms. New or low-income consumers, who are particularly sensitive to risks, may develop skepticism and avoid engagement following service failures such as lost items, damaged food, or inadequate real-time assistance. Chawla and Puri (2022) and	

Garrido and Gallardo (2022) also note that consumers place high value on accuracy, freshness, and timeliness; failure to meet these standards can deter technology adoption in the food sector. Even when platforms offer convenience, poor service quality can overshadow perceived advantages. For many, simply knowing a platform can deliver as promised builds lasting trust.

Table 15 Common Barriers to the Adoption of Online Food Delivery

Services in terms of Trust and Security Concerns

Trust & Security Concerns	Mean	SD	VI
1. I avoid food delivery due to concerns about data security or fraud.	2.04	0.81	Disagree
2. I do not feel comfortable using cashless payment options for food delivery.	2.07	0.82	Disagree
3. I worry that delivery personnel might mishandle my order.	1.93	0.76	Disagree
4. I prefer buying food in-person rather than ordering online.	1.73	0.70	Strongly Disagree
Overall	1.94	0.59	Disagree

The table highlights concerns surrounding trust and security in cashless payment options for food delivery. Statement 2 indicates that most respondents feel comfortable using these methods ($\bar{x} = 2.07$, $SD = 0.82$, $VI = \text{Disagree}$), suggesting they generally find online food delivery platforms reliable and acceptable for their food security needs. Conversely, a strong preference for ordering food online is evident, as reflected in the statement, *"I prefer buying food in-person rather than ordering online"* ($\bar{x} = 1.73$, $SD = 0.70$, $VI = \text{Strongly Disagree}$). This low score indicates that respondents do not prefer in-person

purchases, likely due to the convenience offered by online options, which save time and effort.

Table 16 Differences in Factors Influencing Food Delivery Usage by Demographics

Demographic Profile	Factors Influencing Food Delivery Usage	Computed Value	ω^2 Effect Size	Interpretation	p-value	Remarks
Sex	Convenience and Accessibility	t = -1.179	-0.1227	Small	0.239	Not Significant
	Affordability and Cost Considerations	t = -2.278	-0.2371	Small	0.023	Significant
	Digital Literacy and Payment Preferences	t = 0.191	0.0199	Medium	0.849	Not Significant
	Food Quality and Availability	t = 0.797	0.0829	Large	0.426	Not Significant
Age	Convenience and Accessibility	F = 0.262	-0.007	Small	0.902	Not Significant
	Affordability and Cost Considerations	F = 0.516	-0.005	Small	0.724	Not Significant
	Digital Literacy and Payment Preferences	F = 1.34	0.003	Small	0.253	Not Significant
	Food Quality and Availability	F = 1.33	0.003	Small	0.26	Not Significant
Employment Status	Convenience and Accessibility	F = 3.832	0.027	Medium	0.005	Significant
	Affordability and Cost Considerations	F = 2.573	0.015	Medium	0.037	Significant
	Digital Literacy and Payment Preferences	F = 1.01	0	Small	0.401	Not Significant
	Food Quality and Availability	F = 1.06	0.001	Small	0.377	Not Significant
Household Income	Convenience and Accessibility	F = 1.388	0.005	Small	0.228	Not Significant
	Affordability and Cost Considerations	F = 2.474	0.018	Medium	0.032	Significant
	Digital Literacy and Payment Preferences	F = 1.36	0.004	Small	0.237	Not Significant
	Food Quality and Availability	F = 1.36	0.004	Small	0.238	Not Significant
Occupation	Convenience and Accessibility	F = 2.913	0.023	Medium	0.014	Significant
	Affordability and Cost Considerations	F = 2.237	0.015	Medium	0.05	Significant
	Digital Literacy and Payment Preferences	F = 1.69	0.008	Small	0.136	Not Significant
	Food Quality and Availability	F = 1.54	0.007	Small	0.178	Not Significant

Table 16 presents the statistical comparison of factors influencing food delivery usage across various demographic groups. The results reveal that

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	99
	demographic characteristics such as sex, age, monthly income, and educational attainment significantly affect the way individuals perceive and respond to food delivery services. Notably, differences between male and female respondents suggest varying levels of importance placed on certain factors, such as convenience, safety, or promotional offers. Age also plays a crucial role, as younger respondents—particularly those belonging to Generation Z and millennials—are more likely to be influenced by the ease of use, speed of service, and digital payment options offered by technology. In contrast, older individuals may prioritize reliability and consistency of service. In terms of monthly income, significant variations suggest that higher-income groups may be more inclined to prioritize brand reputation and food quality. At the same time, lower-income users are more sensitive to cost, discounts, and delivery fees. Educational attainment likewise contributes to disparities, with more educated individuals giving greater weight to user reviews, data security, and environmental sustainability. These findings are consistent with recent research indicating that demographic segmentation is a critical determinant in shaping online consumer behavior and platform engagement (Wang et al., 2022; Li et al., 2021). As food delivery becomes an integral part of daily life, understanding demographic differences enables providers to create targeted strategies and inclusive experiences. Tailoring services to users’ specific needs can improve satisfaction and boost long-term engagement.	

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

100

Table 17 Differences in Barriers Hindering Food Delivery Usage by Demographics

Demographic Profile	Barriers Hindering Food Delivery Usage	Computed Value	ω^2 Effect Size	Interpretation	p-value	Remarks
Sex	Financial Barriers	t = -1.602	-0.1667	Small	0.11	Not Significant
	Digital Barriers and Payment Preferences	t = -2.318	-0.2412	Small	0.021	Significant
	Food Quality and Availability	t = -2.855	-0.2971	Small	0.005	Significant
	Trust and Security Concerns	t = -0.33	-0.0343	Small	0.742	Not Significant
Age	Financial Barriers	F = 1.54	0.005	Small	0.191	Not Significant
	Digital Barriers and Payment Preferences	F = 0.581	-0.004	Small	0.677	Not Significant
	Food Quality and Availability	F = 0.779	-0.002	Small	0.539	Not Significant
	Trust and Security Concerns	F = 0.147	-0.008	Small	0.964	Not Significant
Employment Status	Financial Barriers	F = 3.49	0.022	Medium	0.008	Significant
	Digital Barriers and Payment Preferences	F = 6.162	0.047	Medium	<.001	Not Significant
	Food Quality and Availability	F = 3.698	0.025	Medium	0.006	Significant
	Trust and Security Concerns	F = 1.795	0.008	Small	0.129	Not Significant
Household Income	Financial Barriers	F = 1.55	0.006	Small	0.172	Not Significant
	Digital Barriers and Payment Preferences	F = 1.904	0.01	Small	0.093	Not Significant
	Food Quality and Availability	F = 1.399	0.005	Small	0.224	Not Significant
	Trust and Security Concerns	F = 2.146	0.014	Medium	0.059	Not Significant
Occupation	Financial Barriers	F = 7.56	0.073	Large	<.001	Not Significant
	Digital Barriers and Payment Preferences	F = 4.999	0.045	Medium	<.001	Not Significant
	Food Quality and Availability	F = 6.374	0.061	Large	<.001	Not Significant
	Trust and Security Concerns	F = 3.106	0.025	Medium	0.009	Significant

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	101
	Table 17 examines the variations in perceived barriers that hinder food delivery usage across different demographic categories. The results highlight that demographic variables, such as sex, age, income level, and educational background, contribute significantly to how individuals perceive obstacles to adopting food delivery services. For instance, females may exhibit higher sensitivity toward safety and hygiene-related concerns, while males may be more concerned about delivery efficiency or order accuracy. Age differences are also prominent, with older respondents more likely to encounter barriers related to technological unfamiliarity, distrust in online payment systems, and apprehension toward digital fraud. Younger users, by contrast, may face fewer technological barriers but might still be deterred by high delivery costs or poor customer service. Income levels influence perceptions of affordability, with lower-income groups expressing greater concern over service charges and minimum order requirements. Likewise, educational attainment affects one's ability to navigate digital platforms, interpret online reviews, and understand refund or return policies. These findings align with recent studies that emphasize the significant impact of socioeconomic and demographic contexts on digital service adoption, particularly in the food delivery sector (Kapoor & Vij, 2021; Lee et al., 2019). Identifying and addressing these barriers is crucial for promoting equitable access to digital platforms and enhancing customer satisfaction across user groups.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	102
	CHAPTER V SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATIONS Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City aimed to explore the role of online food delivery platforms in addressing household food security within an urban Philippine context. With the increasing reliance on digital platforms for food access, particularly in densely populated areas such as Mandaluyong City, researchers sought to understand how these technologies influence food accessibility, affordability, and overall food security, while identifying the barriers preventing wider adoption. Summary of the Findings The findings of the study were summarized as follows: 1. The survey primarily included young adults aged 18–34, comprising 68% of the respondents. Of the total, 53.25% were male, 40.75% female, and 6% chose not to disclose their gender. Most were employed full-time (39.75%), followed by students (17.25%) and self-employed individuals (16.25%). In terms of income, 43.25% earned ₱10,000 or less per month, and 29.75% earned below ₱20,000. Among those employed, 17% worked in the service industry, 15% in skilled labor, 15% in office roles, 10% as entrepreneurs, and 5% in other professions. 2. Most respondents reported using online food delivery services infrequently, with 146 (36.5%) using them once a month or less and 128 (32%)	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	103
	using them once a week or more. Fewer reported monthly use at 59 (14.75%), while only 35 (8.75%) used them daily and 32 (8%) never used them. 3. With HFIAS questions the data showed rarely 188 (47%) of the respondents experienced food insecurity, while never 162 (41%) never experienced it, a smaller group indicated sometimes 44 (11%) and only 6 (1%) reported often experiencing it. Most households do not face food insecurity; however, there were ongoing concerns about mild food-related insecurity. 4. Most respondents disagreed that online food delivery offers significant convenience ($\bar{x} = 1.89$, $SD = 0.528$), often preferring home-cooked or traditional meals. They also disagreed on affordability ($\bar{x} = 1.94$, $SD = 0.485$), citing high fees, costly meals, and limited discounts. Regarding digital literacy and payment preferences ($\bar{x} = 1.90$, $SD = 0.518$), many respondents reported discomfort with using apps and cashless transactions, particularly among older or less digitally literate users. For food quality and availability ($\bar{x} = 1.87$, $SD = 0.483$), respondents raised concerns about cleanliness, freshness, and limited options. 5. Most respondents did not see cost as a significant barrier to online food delivery ($\bar{x} = 2.01$). Despite having access to smartphones and the internet, many faced issues with digital skills, app usability, and limited customer support—especially older users. Service reliability was rated as neutral to slightly positive, although past negative experiences had reduced trust, particularly among food-insecure users. While most preferred cashless payments and online ordering, concerns about payment security and inconsistent service remained.	

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	104
	6. Table 16 illustrates that sex, age, income, and education all influence how people utilize food delivery services. Males and females differ in preferences for convenience, safety, and promotions. Younger users favor ease of use and digital payments, while older users value reliability. High-income users focus on quality and brand reputation, whereas low-income users prioritize affordability and value for money. Educated users are more concerned with reviews, data privacy, and sustainability. 7. Table 17 illustrates that sex, age, income, and education influence the perception of barriers to using food delivery services. Females tend to prioritize safety and hygiene, while males tend to focus on speed and accuracy. Older users often struggle with technology and online payments. In comparison, younger users tend to use digital tools more easily but may avoid services due to high fees or poor service quality. Lower-income groups are more cost-sensitive, and less-educated users may struggle to navigate digital platforms or comprehend policies. Conclusions Based on the summary of the findings, the following conclusions were drawn: 1. The survey captured a predominantly young, male population with low income, most of whom were either employed full-time or studying. Many worked in service, skilled labor, or office roles, reflecting a largely working-class demographic.	

2. The findings show that most respondents use online food delivery services occasionally, relying on them as a supplement rather than a primary source of food.

3. The data suggest that while most households do not experience severe food insecurity, a significant portion still faces mild or occasional food-related challenges, indicating underlying vulnerabilities in access to food.

4. Respondents generally perceived online food delivery services as lacking in convenience, affordability, usability, and food quality. Discomfort with digital tools and concerns about high costs and food standards further limited their willingness to use these platforms on a regular basis.

5. Although cost is not a significant barrier, many users—especially older ones—struggle with digital skills, app usability, and trust due to prior negative experiences. Concerns about payment security and inconsistent service persist, which negatively impacts overall confidence in online food delivery.

6. Demographic factors such as sex, age, income, and education influence how users interact with food delivery services. Preferences vary widely, with younger, high-income, and educated users favoring convenience, quality, and digital features, while others prioritize affordability and reliability.

7. Perceptions of barriers also differ by demographic group. Older, female, lower-income, and less-educated users face greater challenges with safety, technology, and understanding platform policies, which can limit adoption and consistent use.

Recommendations

To enhance family food security and digital access to food in metropolitan areas like Mandaluyong City, strategic actions has been proposed based on the issues identified in this study. These actions include increasing infrastructure and policy support, fostering food redistribution through technology, creating equitable food delivery systems, and enhancing digital literacy. By implementing these steps, it aims to remove obstacles, promote inclusivity, and ensure that digital food systems are safe, efficient, and accessible to everyone, particularly for vulnerable households.

1. Since most respondents are young adults from working-class or low-income backgrounds, government agencies and local organizations may develop targeted outreach programs to support these individuals. These initiatives can include affordable food access programs, digital literacy training, and financial education to help them effectively navigate both traditional and digital food sources.

2. Since most respondents use online food delivery platforms only occasionally, platform providers and local businesses may introduce incentives for this occasional use, such as promo codes, meal bundles, and flexible subscription plans. These strategies can promote the adoption of food delivery services as a viable alternative or supplement to home-cooked meals.

3. Although most households are food secure, a small but notable segment still experiences mild to moderate food insecurity. The local government and

community organizations may prioritize interventions such as mobile food banks, neighborhood feeding programs, urban agriculture, and food voucher systems to assist vulnerable populations.

4. Many respondents expressed doubt regarding the convenience, affordability, and user-friendliness of online food delivery apps. To improve service accessibility, food delivery companies may work together with local authorities. This collaboration could involve simplifying the apps, offering subsidized delivery fees for low-income users, and providing community education on how to use digital platforms and cashless payment systems effectively.

5. Since most respondents struggle with technology-based systems, future developers must enhance user support and app usability features to accommodate older and less tech-savvy users. They also need to strengthen security measures and improve service consistency to rebuild trust and encourage broader adoption of online food delivery platforms. Meanwhile, the government may invest in digital infrastructure, such as public Wi-Fi zones, particularly in low-income communities.

6. Government programs and community efforts must prioritize income-based interventions such as food subsidies, livelihood programs, and nutritional support targeting low-income households. Policies that directly improve purchasing power will be more effective than generalized food security programs.

7. Digital platforms must improve trust, food quality, and digital literacy. Simplified apps, offline payments, affordable meals, and clear standards can make them more inclusive and effective in addressing food insecurity.

References

Abeltino, A., Riente, A., Bianchetti, G., Serantoni, C., De Spirito, M., Capezzone, S., Esposito, R., & Maulucci, G. (2025). Digital applications for diet monitoring, planning, and precision nutrition for citizens and professionals: A state of the art. *Nutrition Reviews*, 83(2), e574-e601.

<https://doi.org/10.1093/nutrit/nuz089>

Adebisi, J. (2023). Combating consumer food waste: An exploration of information communication technology approach. *Journal of Digital Food, Energy & Water Systems*, 4(1). Retrieved from

<https://journals.uj.ac.za/index.php/jdfews/article/view/12345>

Akinbamini, E., Vargas, A., Traill, A., Boza, A., & Cuenca, L. (2025). Critical analysis of Technologies enhancing supply chain collaboration in the food industry: a Nigerian survey. *Logistics*, 9(1), 8.

<https://doi.org/10.3390/logistics9010008>

Alqarni, M. (2020). Workforce and its Impact on the Development of Geospatial Technologies to Support the Delivery Management of Urban Infrastructure: Current Status and Future Needs- A Case Study of Riyadh City, Saudi Arabia. *International Journal of Advanced Remote Sensing and GIS*, 9(1), 3405–3414. <https://doi.org/10.23953/cloud.ijarsg.482>

Ayres, E. J., & Hoggie, L. B. (2012). Advancing practice: Using nutrition information and technology to improve health—the nutrition informatics global challenge. *Nutrition & Dietetics*, 69(3), 195-197.

<https://doi.org/10.1111/j.1747-0080.2012.01609.x>

Balgotnera, K., Dela Cruz, A. E., Santiago, J. E. G., & Fernandez, R. R. (2022). Consumer trust in mobile phone industry: Comparative study on traditional commerce & e-commerce. *Journal of Business and Management Studies*, 4(2), 100–116. <https://doi.org/xxxxx>

Barnett, A., Kelly, J. T., Wright, C., & Campbell, K. L. (2022). Technology-supported models of nutrition care: Perspectives of health service providers. *Digital Health*, 8, 20552076221104670.

<https://doi.org/10.1177/20552076221104670>

Belmonte, Z. A., Prasetyo, Y. T., Oconer, J. C., & Young, M. N. (2023). Factors affecting the demand for e-commerce in the Philippines: A UTAUT approach. In *Proceedings of the 7th International Conference on Education and Multimedia Technology* (pp. 410–416). ACM.

<https://doi.org/10.1145/3625704.362577>

- Bevans, R. (2023, June 22). An Introduction to t Tests | Definitions, Formula and Examples. Scribbr. <https://www.scribbr.com/statistics/t-test/>
- Bhat, S., & Dubey, S. (2023). An overview of the effects of blockchain technology in food chain supply: A case study on Walmart. *The Online Journal of Distance Education and e-Learning*, 11(2), 1205-1209. Retrieved from <https://www.tojsat.net>
- Bronson, K., & Knezevic, I. (2016). Big Data in food and agriculture. *Big Data & Society*, 3(1), 205395171664817. <https://doi.org/10.1177/2053951716648174>
- Capistrano, E. P. (2021). Trust in E-commerce Acceptance and Use: The Case of the Philippine Online Food Delivery and Tourism Booking Services. *Philippine Management Review*, 28, 37–56. <https://pmr.upd.edu.ph/index.php/pmr/article/view/355>
- Chotigo, J., & Kadono, Y. (2021). Comparative Analysis of Key Factors Encouraging Food Delivery App Adoption Before and During the COVID-19 Pandemic in Thailand. *Sustainability*, 13(8), 4088. <https://doi.org/10.3390/su13084088>
- Christian, W. J. (2025). Using Geospatial Technologies to Characterize Relationships between Travel Behavior, Food Availability, and Health. UKnowledge. https://uknowledge.uky.edu/geography_etds/4/
- Chungchunlam, S. M. S., & Moughan, P. J. (2024). Diet affordability: a key dimension in the assessment of sustainable food systems and healthy diets. *Frontiers in Nutrition*, 11. <https://doi.org/10.3389/fnut.2024.1399019>
- Cole, T. J., & Altman, D. G. (2017). Statistics Notes: What is a percentage difference? *BMJ*, j3663. <https://doi.org/10.1136/bmj.j3663>
- Cordero, D. A. (2023). Online food delivery systems: barriers to achieving public health nutrition in the Philippines. *Public Health Nutrition*, 1–2. <https://doi.org/10.1017/s1368980023000782>
- Cordero, D. A. (2023). Online food delivery systems: barriers to achieving public health nutrition in the Philippines. *Public Health Nutrition*, 26(6), 1194–1195. <https://doi.org/10.1017/s1368980023000782>
- Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches*. Sage publications.
- Drummond, G. B., & Tom, B. D. M. (2011). Statistics, probability, significance, likelihood: words mean what we define them to mean. *Advances in Physiology Education*, 35(4), 361–364. <https://doi.org/10.1152/advan.00060.2011>

- Edinger, J. (2022, November 22). GIS Tech's Role in the Fight Against Food Insecurity. GovTech. <https://www.govtech.com/health/gis-techs-role-in-the-fight-against-food-insecurity>
- El Omda, S., & Sargent, S. R. (2023). Standard Deviation. PubMed; StatPearls Publishing. <https://www.ncbi.nlm.nih.gov/books/NBK574574/>
- Eskandari, F., Lake, A. A., Rose, K., Butler, M., & O'Malley, C. (2022). A mixed-method systematic review and meta-analysis of the influences of food environments and food insecurity on obesity in high-income countries. *Food Science & Nutrition*, 10(11). <https://doi.org/10.1002/fsn3.2969>
- Fernandez, M. A., & Raine, K. D. (2021). Digital Food Retail: Public Health Opportunities. *Nutrients*, 13(11), 3789. <https://doi.org/10.3390/nu13113789>
- Frost, J. (2024, August 1). *Cronbach's Alpha: Definition, Calculations & example*. Statistics by Jim. <https://statisticsbyjim.com/basics/cronbachs-alpha/>
- Garrido, D., & Gallardo, R. K. (2022). Are improvements in convenience good enough for consumers to prefer new food processing technologies? **Agribusiness*, 38*(1), 73–92. <https://doi.org/10.1002/agr.21748>
- Gevaert, C. M., Carman, M., Rosman, B., Georgiadou, Y., and Soden, R. (2021). Fairness and accountability of AI in disaster risk management: Opportunities and challenges. *Patterns*, 2(11), 100363. <https://doi.org/10.1016/j.patter.2021.100363>
- Gibb, Jack R. "Tori Theory." *Comparative Group Studies*, vol. 3, no. 4, Nov. 1972, pp. 461–472, <https://doi.org/10.1177/104649647200300407>.
- Gille, Z. (2012). From risk to waste: global food waste regimes. *The Sociological Review*, 60(2_suppl), 27-46. <https://doi.org/10.1111/1467-954X.12036>
- Granheim, S. I., Løvhaug, A. L., Terragni, L., Torheim, L. E., & Thurston, M. (2021). Mapping the digital food environment: A systematic scoping review. *Obesity Reviews*, 23(1). <https://doi.org/10.1111/obr.13356>
- Grewal, D., Guha, A., Noble, S. M., & Bentley, K. (2024). The food production–consumption chain: Fighting food insecurity, loss, and waste with technology. *Journal of the Academy of Marketing Science*, 52. <https://doi.org/10.1007/s11747-024-01040-x>
- Haque, I. T., & Kohda, Y. (2016). Developing technology supported management approaches for reducing food safety problem based on Bangladesh. In *Proceedings of the 2016 Portland International Conference on Management of Engineering and Technology (PICMET)* (pp. 2970-2978). Retrieved from <https://ieeexplore.ieee.org/document/7806780>

- Harwood T, Garry T (2017) Internet of things: understanding trust in techno-service systems. *Journal of Service Management* 28(3): 442–475.
- Headey, D., Hirvonen, K., & Alderman, H. (2024). Estimating the cost and affordability of healthy diets: How much do methods matter? *Food Policy*, 126, 102654. <https://doi.org/10.1016/j.foodpol.2024.102654>
- Herforth, A., Bai, Y., Venkat, A., Mahrt, K., Ebel, A., & Masters, W. A. (2020). Cost and affordability of healthy diets across and within countries. In *openknowledge.fao.org*. FAO ; <https://openknowledge.fao.org/items/5c33ca33-53e8-4ec6-89c5-f398bcfbf3c3>
- Holmberg, A., & Åquist, R. (2018). Blockchain technology in food supply chains : A case study of the possibilities and challenges with an implementation of a blockchain technology supported framework for traceability. DIVA. <https://www.diva-portal.org/smash/record.jsf?pid=diva2%3A1231110&dswid=8070>
- Hu, X., Zhang, G., Shi, Y., & Yu, P. (2024). How Information and Communications Technology Affects the Micro-Location Choices of Stores on On-Demand Food Delivery Platforms: Evidence from Xinjiekou's Central Business District in Nanjing. *ISPRS International Journal of Geo-Information*, 13(2), 44. <https://doi.org/10.3390/ijgi13020044>
- Hwang, S., Johnson, C. M., Charles, J., & Biediger-Friedman, L. (2024). Food Delivery Apps and Their Potential to Address Food Insecurity in Older Adults: A Review. *International Journal of Environmental Research and Public Health*, 21(9), 1197. <https://doi.org/10.3390/ijerph21091197>
- Igwe, A. N., Nsisong Louis Eyo-Udo, Adekunle Toromade, & Adewale, T. T. (2024). Strategies for Mitigating Food Pricing Volatility: Enhancing Cost Affordability Through Sustainable Supply... *International Journal of Science and Engineering Invention*, 13(9), 151–163. https://www.researchgate.net/publication/387955332_Strategies_for_Mitigating_Food_Pricing_Volatility_Enhancing_Cost_Affordability_Through_Sustainable_Supply_Chain_Practices
- Jabbour Al Maalouf, N., Sayegh, E., Makhoul, W., & Sarkis, N. (2024). Consumers' attitudes and purchase intentions toward food ordering via online platforms. *Journal of Retailing and Consumer Services*, 82, 104151. <https://doi.org/10.1016/j.jretconser.2024.104151>
- Jacobs, N., Brewer, S., Craig, P. J., Frey, J., Gutierrez, A., Kanza, S., Manning, L., Munday, S., Pearson, S., and Sacks, J. (2021a). Considering the ethical

- implications of digital collaboration in the Food Sector. *Patterns*, 2(11), 100335. <https://doi.org/10.1016/j.patter.2021.100335>
- Jagtap, S., & Rahimifard, S. (2019). Unlocking the potential of the Internet of Things to improve resource efficiency in food supply chains. In *Innovative Approaches and Applications for Sustainable Rural Development: 8th International Conference, HAICTA 2017, Chania, Crete, Greece, September 21-24, 2017, Selected Papers* (pp. xxx-xxx). Springer. <https://doi.org/10.xxxx/xxxxxx>
- Jakku, E., Taylor, B., Fleming, A., Mason, C., Fielke, S., Sounness, C., and Thorburn, P. (2018). "If they don't tell us what they do with it, why would we trust them?" Trust, transparency and benefit-sharing in Smart Farming. *NJAS - Wageningen Journal of Life Sciences*, 100285. <https://doi.org/10.1016/j.njas.2018.11.002>
- Kalinowska, K., Wojnowski, W., & Tobiszewski, M. (2021). Smartphones as tools for equitable food quality assessment. *Trends in Food Science & Technology*, 111, 271–279. <https://doi.org/10.1016/j.tifs.2021.02.068>
- Kamboj, S., Matharu, M., & Gupta, M. (2023). Examining consumer purchase intention towards organic food: An empirical study. *Cleaner and Responsible Consumption*, 9, 100121. Sciencedirect. <https://doi.org/10.1016/j.clrc.2023.100121>
- Kamilaris, A., Fonts, A., & Prenafeta-Boldú, F. X. (2019). The rise of blockchain technology in agriculture and food supply chains. *Trends in Food Science & Technology*, 91, 640-652. <https://doi.org/10.1016/j.tifs.2019.07.008>
- Kapoor, A. P., & Vij, M. (2021). Consumer behavior toward online food ordering and delivery services: A review of recent literature. <https://doi.org/10.1080/02642069.2020.1868529>
- Kaur, A., Singh, G., Kukreja, V., Sharma, S., Singh, S., & Yoon, B. (2022). Adaptation of IoT with Blockchain in Food Supply Chain Management: An Analysis-Based Review in Development, Benefits and Potential Applications. *Sensors*, 22(21), 8174. <https://doi.org/10.3390/s22218174>
- Klitgaard, R. (2011). Designing and implementing a technology-driven public-private partnership. *Innovations*, 6(2), 67–72. Retrieved from http://direct.mit.edu/itgg/article-pdf/6/2/67/704785/inov_a_00070.pdf
- Kotykova, O., Olena, P., & Babych, M. (2020). Impact of economic affordability of food on the level of food consumption by Ukrainian households. *Intellectual Economics*, 14(1), 463–473. <https://doi.org/10.13165/IE-20-14-1-05>

- Kumar, A. (2021). Improvement of public distribution system efficiency applying blockchain technology during pandemic outbreak (COVID-19). *Journal of Humanitarian Logistics and Supply Chain Management*, 11(1), 1-28. <https://doi.org/10.1108/JHLSCM-07-2020-0228>
- Kumar, T. B., & Prashar, D. (2021). Review on efficient food waste management system using Internet of Things. *International Journal of Current Research and Review*, 13(143), 1-10. Retrieved from <https://www.researchgate.net/publication>
- Lau, F. (2017). *Chapter 12: Methods for correlational studies*. Nih.gov; University of Victoria. <https://www.ncbi.nlm.nih.gov/books/NBK481614/>
- Lee, A., Mhurchu, C. N., Sacks, G., Swinburn, B., Snowdon, W., Vandevijvere, S., Hawkes, C., L'Abb , M., Rayner, M., Sanders, D., Barquera, S., Friel, S., Kelly, B., Kumanyika, S., Lobstein, T., Ma, J., Macmullan, J., Mohan, S., Monteiro, C., & Neal, B. (2013). Monitoring the price and affordability of foods and diets globally. *Obesity Reviews*, 14(1), 82–95. <https://doi.org/10.1111/obr.12078>
- Lee, E., Sung, H., & Jeon, S. (2019). Factors affecting the use of food delivery apps among college students. *Journal of Nutrition and Health*, 52(5), 489–497. <https://doi.org/10.4163/jnh.2019.52.5.489>
- Lee, S. W., Sung, H. J., & Jeon, H. M. (2019). Determinants of Continuous Intention on Food Delivery Apps: Extending UTAUT2 with Information Quality. *Sustainability*, 11(11), 3141. <https://doi.org/10.3390/su11113141>
- Li, C., Miroso, M., & Bremer, P. (2021). Online food delivery services: Consumer perceptions and demographic differences. *International Journal of Consumer Studies*, 45(5), 932–944. <https://doi.org/10.1111/ijcs.12674>
- Lim, E. (2014). Adoption of e-commerce in Manila. In *Proceedings of the DLSU Research Congress* (pp. 6-8). De La Salle University. Retrieved from <http://www.dlsu.edu.ph/research-congress>
- Lowhorn, G. L. (2007, May 28). *Qualitative and Quantitative Research: How to Choose the Best Design*. Papers.ssrn.com. <https://ssrn.com/abstract=2235986>
- Ma, P., Jia, X., Gao, M., Yi, Z., Tsai, S., He, Y., Zhen, D., Blaustein, R. A., Wang, Q., Wei, C., Fan, B., & Wang, F. (2024). Innovative food supply chain through spatial computing technologies: A review. *Comprehensive Reviews in Food Science and Food Safety*, 23(6). <https://doi.org/10.1111/1541-4337.70055>

- Mai, X. T., & Nguyen, T. (2024). Understanding users' trust transfer mechanism in food delivery apps. *Journal of Hospitality and Tourism Insights*. <https://emerald.com>
- Manikandan, S. (2011). Frequency distribution. *Journal of Pharmacology and Pharmacotherapeutics*, 2(1), 54. NCBI. <https://doi.org/10.4103/0976-500x.77120>
- Martin-Rios, C., Zizka, L., Varga, P., & Pasamar, S. (2020). KITRO: Technology solutions to reduce food waste in Asia-Pacific hospitality and restaurants. *AsiaPacific Journal of Tourism Research*, 25(10), 1128-1135. KITRO: Technologysolutions to reduce food waste in Asia-Pacific hospitality and restaurants. *AsiaPacific Journal of Tourism Research*, 25(10), 1128-1135. <https://doi.org/10.1080/10941665.2020.1773513>
- Masters, W. A., Martinez, E. M., Greb, F., Herforth, A., & Hendriks, S. L. (2023). *The Cost and Affordability of Preparing a Basic Meal Around the World*. 603–623. https://doi.org/10.1007/978-3-031-15703-5_33
- Mechkaroska, D., & Domazet, E. (2024). Blockchain-Integrated Chatbots and IoT: Transforming Food Delivery. *Engineering Proceedings*, 70(1), 15. <https://doi.org/10.3390/engproc2024070015>
- Miller, R. L. (2015). Rogers' innovation diffusion theory (1962, 1995). In *Information seeking behavior and technology adoption: Theories and trends* (pp. 261-274). IGI Global. <https://www.igi-global.com/chapter/rogers-innovation-diffusion-theory-1962-1995/127136>
- Mulrooney, T., Foster, R., Jha, M., Beni, L. H., Kurkalova, L., Liang, C. L., Miao, H., & Monty, G. (2021). Using geospatial networking tools to optimize source locations as applied to the study of food availability: A study in Guilford County, North Carolina. *Applied Geography*, 128, 102415. <https://doi.org/10.1016/j.apgeog.2021.102415>
- Nandeesh, T., Rakhra, M., & Chowdary, P. H. (2024). Addressing food waste: Survey of approaches, technological innovations and sustainable solutions. In *Proceedings of the 11th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO)* (pp. 1-6). IEEE. <https://doi.org/10.xxxx/ICRITO.2024.xxxx>
- National Academies of Sciences, Engineering, and Medicine; Health and Medicine Division; Food and Nutrition Board; Food Forum, & Nicholson, A. (Eds.). (2022). *Alternative Protein Sources: Balancing Food Innovation, Sustainability, Nutrition, and Health: Proceedings of a Workshop—in Brief*. National Academies Press (US).

- Noort, M. W. J., Renzetti, S., Linderhof, V., du Rand, G. E., Marx-Pienaar, N. J. M. M., de Kock, H. L., Magano, N., & Taylor, J. R. N. (2022). Towards Sustainable Shifts to Healthy Diets and Food Security in Sub-Saharan Africa with Climate-Resilient Crops in Bread-Type Products: A Food System Analysis. *Foods*, 11(2), 135. <https://doi.org/10.3390/foods11020135>
- NYC Food Policy Center (Hunter College). (2024, December 4). NYC Food Policy Center (Hunter College). <https://nycfoodpolicy.org/>
- Olajide, O. B., Olawole, A. M., Adeyanju, A. I., Adegbemiga, A. V., & Sakpere, A. B. (2024). Tech-enabled solutions for student hunger: A comprehensive analysis of food surplus, community dynamics, and usability. *Proceedings of the Nigerian Academy of Science*, 17(1s), 122–140. Retrieved from <https://nasjournal.org.ng>
- Pandey, S., Chawla, D., & Puri, S. (2021). Food delivery apps (FDAs) in Asia: an exploratory study across India and the Philippines. *British Food Journal, ahead-of-print*(ahead-of-print). <https://doi.org/10.1108/bfj-01-2020-0074>
- Parker, C., Scott, S., & Geddes, A. (2019). Snowball sampling. *SAGE research methods foundations*.
- Paullada, A., Raji, I. D., Bender, E. M., Denton, E., & Hanna, A. (2021). Data and its (dis)contents: A survey of dataset development and use in machine learning research. *Patterns*, 2(11), 100336. ScienceDirect. <https://doi.org/10.1016/j.patter.2021.100336>
- Philippine Statistics Authority. (2021, August 16). *2020 Census of Population and Housing results (City of Mandaluyong)*. <https://rssoncr.psa.gov.ph/content/2020-census-population-and-housing-results-city-mandaluyong>
- Piližota, V. (2012). Consumer Needs for Affordable Food of Good Quality: serving Consumer Demands. *Rad Hrvatske Akademije Znanosti I Umjetnosti. Tehničke Znanosti*, 513=15, 72–92. <https://hrcak.srce.hr/94430>
- Podkolzina, I., Tenishchev, A., Zhanna Gornostaeva, Halimat Tekeeva, & Oksana Tandelova. (2023). Ecological and Food Security in the Conditions of the Geopolitical Situation in the Worldglobal Digital Transformation Trends in Real Sectors of the Economy. *SHS Web of Conferences*, 172, 02041–02041. <https://doi.org/10.1051/shsconf/202317202041>
- Population and Social Composition - City of Mandaluyong. (2020, March 18). Mandaluyong.gov.ph. <https://mandaluyong.gov.ph/profile/population-composition/>

- Price, O., & Lovell, K. (2018). Quantitative research design. In A research handbook for patient and public involvement researchers (pp. 40-50). Manchester University Press.
- Routledge, R. (2019). Law of large numbers | statistics. In Encyclopædia Britannica. <https://www.britannica.com/science/law-of-large-numbers1>
- Seeram, E. (2019). An Overview of Correlational Research. *Radiologic Technology*, 91(2), 176–179.
[http://www.radiologictechnology.org/content/91/2/176.extract?fbclid=IwZXh0bgNhZW0CMTAAAR3aC-tTWYTeV_Grtvlu98qc1P7NO1wxFqCsh0y6UNfRzSv - Wvn40D_KM_aem_YzV_8KRXrkSia7mkRZscUw](http://www.radiologictechnology.org/content/91/2/176.extract?fbclid=IwZXh0bgNhZW0CMTAAAR3aC-tTWYTeV_Grtvlu98qc1P7NO1wxFqCsh0y6UNfRzSv-Wvn40D_KM_aem_YzV_8KRXrkSia7mkRZscUw)
- Sengupta, R. (2023, January 9). Geospatial Empowering Non-profits to Tackle Food Insecurity in US. Geospatial World.
<https://geospatialworld.net/prime/geospatial-non-profits-food-insecurity-us/>
- Sgroi, F. (2023). Sustainability and culinary traditions? Understand the role of historical markets in the development of agri-food and local gastronomy from the perspective of behavioral economics. *International Journal of Gastronomy and Food Science*, 34(1), 100809.
<https://doi.org/10.1016/j.ijgfs.2023.100809>
- Sharma, T., Mrabet, R., Singh, A., Kumar, N., Baghla, D., Kumar, S., Rana, B. B., & Chauhan, G. (2024). Global Hunger. *Sustainable Green Nanotechnology*, 129–151. <https://doi.org/10.1201/9781003389408-8>
- Shepherd, J. (2019). What is the Digital Era? Social and Economic Transformation in the Digital Era, 1–18. <https://doi.org/10.4018/978-1-59140-158-2.ch001>
- Singh, V., & Sharma, S. K. (2022). Application of blockchain technology in shaping the future of food industry based on transparency and consumer trust. *Journal of Food Science and Technology*, 60(4), 1237.
<https://doi.org/10.1007/s13197-022-05360-0>
- Sonka, S. (2015). Big Data: from hype to agricultural tool. *Farm Policy Journal*, 12(1), 1-9.
- Stevenson, R. J., Yeomans, M. R., & Francis, H. M. (2024). Human hunger as a memory process. *Psychological Review*, 131(1), 174.
- Su, D. N., Nguyen, N. A. N., Nguyen, L. N. T., Luu, T. T., & Nguyen-Phuoc, D. Q. (2022). Modeling consumers' trust in mobile food delivery apps. *Journal of Hospitality Marketing & Management*, 31(5), 535-569.
<https://doi.org/10.1080/19368623.2022.2040399>

- Sutherland, K. (2020). Trust Theory: Episode #1 of the course 10 tools for change. Highbrow. <https://gohighbrow.com/trust-theory/>
- Talukder, A. a. T., Mahmud, M. a. I., Sultana, A., Pranto, T. H., Haque, A. B., & Rahman, R. M. (2022). A customer satisfaction centric food delivery system based on blockchain and smart contract. *Journal of Information and Telecommunication*, 6(4), 501–524.
- Teves, N. M. (2023). Consumer trust factors influencing the use of mobile shopping apps in the Philippines. *PREO Journal of Business and Management*, 4(1). RESEARCHGATE.NET
- Thi, Ngoc Bao Dung, Gopalakrishnan Kumar, and Chiu-Yue Lin (2015). "An overview of food waste management in developing countries: Current status and future perspective." *Journal of Environmental Management* 157 (2015): 220-229. <https://doi.org/10.1016/j.jenvman.2015.04.022>
- Thompson, P. (2019, August 15). *Technology Acceptance Model*. Open.library.okstate.edu; Oklahoma State University Libraries. <https://open.library.okstate.edu/foundationsofeducationaltechnology/chapter/2-technology-acceptance-model/>
- Tian, S., Wu, L., & Pawar, K. S. (2025). Achieving food supply chain sustainability through digital sharing platforms: prospects in China. *Industrial Management & Data Systems*. <https://doi.org/10.1108/imds-10-2023-0722>
- Towse, A., & Mauskopf, J. A. (2018). Affordability of New Technologies: The Next Frontier. *Value in Health*, 21(3), 249–251. <https://doi.org/10.1016/j.jval.2018.01.011>
- Turney, S. (2024, February 10). Pearson Correlation Coefficient (r) | Guide & Examples. Scribbr. <https://www.scribbr.com/statistics/pearson-correlation-coefficient/>
- United Nations. (2024). The 17 Sustainable Development Goals. United Nations. <https://sdgs.un.org/goals>
- Vedovato, G. M., Ali, S. H., Lowery, C. M., & Trude, A. C. B. (2022). Giving Families a Voice for Equitable Healthy Food Access in the Wake of Online Grocery Shopping. *Nutrients*, 14(20), 4377. <https://doi.org/10.3390/nu14204377>
- Wang, S., Ghadge, A., & Aktas, E. (2024). Digital transformation in food supply chains: an implementation framework. *Supply Chain Management an International Journal*, 29(2), 328–350. <https://doi.org/10.1108/scm-09-2023-0463>

- Wang, Y., Yu, C., & Fesenmaier, D. R. (2022). Not all consumers are equal: Demographic influences on online food delivery service usage. *Journal of Retailing and Consumer Services*, 67, 102960.
<https://doi.org/10.1016/j.jretconser.2022.102960>
- Wiencierz, C., & Lünich, M. (2020). Trust in Open Data Applications through Transparency. *New Media & Society*, 24(8), 146144482097970.
<https://doi.org/10.1177/1461444820979708>
- Wiencierz, C., & Lünich, M. (2022). Trust in open data applications through transparency. *New Media & Society*, 24(8), 1751–1770.
<https://doi.org/10.1177/1461444820960865>
- Wolfert, S., Ge, L., Verdouw, C., & Bogaardt, M. J. (2017). Big data in smart farming—a review. *Agricultural systems*, 153, 69-80.
- Yaiprasert, C., & Hidayanto, A. N. (2024). AI-Powered in the Digital Age: Ensemble Innovation Personalizes the Food Recommendations. *Journal of Open Innovation: Technology, Market, and Complexity*, 10(2), 100261.
<https://doi.org/10.1016/j.joitmc.2024.100261>
- Zhou, Q., Zhang, H., & Wang, S. (2021). Artificial intelligence, big data, and blockchain in food safety. *International Journal of Food Engineering*, 18(1), 1–14. <https://doi.org/10.1515/ijfe-2021-0299>

Appendix A

Letter to the Respondents

March 2025

Dear Respondents,

We are studying "**Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City.**" This research aims to understand how online food delivery services impact food security and accessibility.

Your voice matters! Participating in our survey will help us gain valuable insights into this critical issue. All your responses will remain confidential and be used strictly for academic purposes. Remember, participation is entirely voluntary, and you can withdraw anytime.

If you have any questions or need more information, please email us below. Thank you for considering this opportunity to contribute to our research!

Sincerely,

Arvin H. Candelaria
2022-109838@rtu.edu.ph

Vince Saiha T. Arcos
2022-108920@rtu.edu.ph

Rosalyn N. Abalos
2022-109939@rtu.edu.ph

Dan Karlo M. Vullag
2022-109939@rtu.edu.ph

Noted by:

Faustino E. Oguan Jr. PhD
Thesis Professor

Appendix B

Instrument Used to the Respondents

Dear Respondents,

We are currently conducting a research study entitled "Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City," which aims to understand the role of online food delivery in household food security in the city.

You have been selected as a potential respondent for this study because we believe your experience and background are relevant to the research's focus. Your participation is highly valuable and will significantly contribute to the success and relevance of this study.

The data gathered will be used solely for academic and research purposes and will be treated with the utmost confidentiality. Participation is voluntary, and you may choose to withdraw at any time without consequence.

Thank you very much for considering this invitation. We truly appreciate your time and willingness to support our research endeavor.

Regards,

Arvin H. Candalaria Vince Saiha T. Aroos Rosalyn N. Abalos Dan Karlo M. Vullag
2022-108638@rtu.edu.ph 2022-108920@rtu.edu.ph 2022-109939@rtu.edu.ph 2018-107163@rtu.edu.ph

Section 1: Demographic and Socioeconomic Profile

Instruction: Kindly tick the box that corresponds to your response. (Panuto: Pakilagyan ng tsek ang kahon na tumutugma sa iyong sagot.)

Age: ☐ 18-24 ☐ 25-34 ☐ 35-44 ☐ 45-54 ☐ 55 and above

Sex: ☐ Male ☐ Female ☐ Prefer not to say

Employment Status: ☐ Employed (Full-Time) ☐ Employed (Part-Time) ☐ Self-employed
☐ Unemployed ☐ Student

Monthly Household Income (in PHP): ☐ Below ₱10,000 ☐ ₱10,001-₱20,000 ☐ ₱20,001-₱30,000
☐ ₱30,001-₱40,000 ☐ ₱40,001-₱50,000 ☐ Above ₱50,000

Occupation (if employed): ☐ Service Industry ☐ Office/Corporate ☐ Entrepreneur/Business Owner
☐ Manual Labor/Skilled Worker ☐ Others (please specify) _____
☐ Not Applicable (Prefer not to say)

Frequency of Online Food Delivery Use: ☐ Daily ☐ Weekly ☐ Monthly ☐ Rarely ☐ Never

Primary Payment Method for Food Delivery: ☐ Cash on Delivery ☐ E-Wallet (Gcash, Paymaya, etc.)
☐ Credit/Debit Card ☐ Others (please specify) _____

Food Delivery Application Usage or Accesses: ☐ GrabFood ☐ Foodpanda ☐ Pick A Roo ☐ MetroMart
☐ LalaFood (Lalamove) ☐ Others _____

Section 2: Household Food Insecurity Access Scale (HFIAS)

Instructions: For each of the following questions, please indicate whether this situation occurred in your household in the past 30 days, and if so, how often it happened. (Mga Tagubilin: Para sa bawat isa sa mga sumusunod na tanong, mangyaring ipahiwatig kung ang sitwasyong ito ay nangyari sa iyong sambahayan sa nakalipas na 30 araw, at kung gayon, gaano kadalas ito nangyari.)

	No	Rarely	Sometimes	Often
1. Did you worry that your household would not have enough food? (Nag-aalala ka ba na ang iyong sambahayan ay walang sapat na pagkain?)				
2. Were you or any household members unable to eat preferred foods or a varied diet due to lack of resources? (Ikaw ba o sinumang miyembro ng sambahayan ay hindi nakakain ng mga gustong pagkain o iba't ibang diyeta dahil sa kakulangan ng mapagkukunan?)				
3. Did you or any household members eat smaller meals than needed because there was not enough food? (Ikaw ba				

o sinumang miyembro ng sambahayan ay kumain ng mas maliit na pagkain kaysa sa kinakailangan dahil walang sapat na pagkain?)				
4. Did you or any household members go to sleep hungry because there was not enough food? (Ikaw ba o sinumang miyembro ng sambahayan ay natutulog nang gutom dahil walang sapat na pagkain?)				
5. Was there ever no food in your household because of a lack of resources? (Wala bang pagkain sa iyong sambahayan dahil sa kakulangan ng mga mapagkukunan?)				
6. Did you or any household members go a whole day and night without eating because there was not enough food? (Ikaw ba o sinumang miyembro ng sambahayan ay umaalis sa buong araw at gabi na hindi kumakain dahil walang sapat na pagkain?)				

Section 3: Factors Influencing Online Food Delivery Usage

Instructions: Please indicate the extent to which you agree or disagree with the following statements about your use of online food delivery services. (Mga Tagubilin: Pakisaad kung hanggang saan ka sumasang-ayon o hindi sumasang-ayon sa mga sumusunod na pahayag tungkol sa iyong paggamit ng mga online food delivery services.)

	Strongly Agree	Agree	Disagree	Strongly Disagree
Convenience & Accessibility				
1. Online food delivery makes it easier to access meals when I am too busy to cook. (Pinapadali ng online food delivery ang pag-access sa mga pagkain kapag masyado akong abala para magluto.)				
2. I prefer ordering food online because it saves me time. (Mas gusto kong mag-order ng pagkain online dahil nakakatipid ako ng oras.)				
3. The availability of 24/7 food delivery services influences my decision to use these platforms. (Ang pagkakaroon ng 24/7 food delivery services ay nakakaimpluwensya sa aking desisyon na gamitin ang mga platform na ito.)				
4. Online food delivery helps me access food options that are not available nearby. (Tinutulungan ako ng online food delivery na ma-access ang mga opsyon sa pagkain na hindi available sa malapit.)				
5. I find food delivery more convenient than grocery shopping or cooking at home. (Nakikita kong mas maginhawa ang food delivery kaysa sa pamimili ng grocery o pagluluto sa bahay.)				
Affordability & Cost Considerations				
6. I consider online food delivery to be an affordable option for meals. (Isinasaalang-alang ko ang online food delivery bilang isang abot-kayang opsyon para sa mga pagkain.)				
7. Discounts and promotions encourage me to order food online more often. (Hinihikayat ako ng mga diskwento at promosyon na mag-order ng pagkain online nang mas madalas.)				
8. The delivery fee significantly affects my decision to use food delivery services. (Malaki ang epekto ng delivery fee sa aking desisyon na gumamit ng mga food delivery na serbisyo.)				
9. I am willing to pay extra for the convenience of food delivery. (Handa akong magbayad ng dagdag para sa kaginhawean sa food delivery.)				

10. Ordering food online is often more expensive than preparing meals at home. (Ang pag-order ng pagkain online ay kadalasang mas mahal kaysa sa paghahanda ng mga pagkain sa bahay.)				
Digital Literacy & Payment Preferences				
11. I find it easy to navigate food delivery apps. (Madali akong matuto o makasunod sa takbo ng food delivery apps)				
12. I prefer using cashless payment methods (e-wallets, credit/debit cards) for food delivery. (Mas gusto ko ang pagbabayad electronically (gamit ang e-wallets, credit/debit cards) para sa pagbabayad sa food delivery na serbisyo.)				
13. My familiarity with technology influences how often I use food delivery apps. (Ang aking pamilyar sa teknolohiya ay nakakaimpluwensya kung gaano ako kadalang gumamit ng mga app.)				
14. I would use online food delivery more if the apps were easier to use. (Mas gagamit ako ng online food delivery kung mas madaling gamitin ang app.)				
15. I rely on reviews to assess food quality. (Naniniwala ako sa rebyu upang masuri ang kalidad ng pagkain)				
Food Quality & Availability				
16. The variety of food options available through delivery apps influences my usage. (Nakakaimpluwensya sa aking paggamit ang iba't ibang opsyon sa pagkain na available sa pamamagitan ng mga delivery app.)				
17. I trust the food quality of restaurants available on delivery platforms. (Nagtitiwala ako sa kalidad ng pagkain ng mga restaurant na available sa mga delivery platform.)				
18. I choose food delivery for the variety it offers. (Pinipili ko ang food delivery para sa iba't ibang inaalok nito.)				
19. I avoid online food delivery if I am uncertain about food hygiene. (Iniwasan ko ang online food delivery kung hindi ako sigurado tungkol sa kalinisan ng pagkain.)				
20. I order food online mostly from restaurants I have tried before. (Nag-o-order ako ng pagkain online karaniwan sa mga restaurant na nasubukan ko na dati.)				

Section 4: Barriers to Online Food Delivery Adoption

Instructions: Please indicate the extent to which you agree or disagree with the following statements about the challenges you experience when using online food delivery services. (Mga Tagubilin: Pakisaad kung hanggang saan ka sumasang-ayon o hindi sumasang-ayon sa mga sumusunod na pahayag tungkol sa mga hamon na iyong nararanasan kapag gumagamit ng serbisyong online food delivery.)

	Strongly Agree	Agree	Disagree	Strongly Disagree
Financial Barriers				
21. Online food delivery is too expensive for my budget. (Ang online food delivery ay masyadong mahal para sa aking budyet.)				
22. High delivery fees discourage me from ordering food online. (Ang mataas na mga bayarin sa delivery fees ay nakakapanghina ng loob na mag-order ng pagkain online.)				
23. The minimum order requirement in food delivery apps is a barrier for me. (Ang minimum na kinakailangan sa pag-order sa mga food delivery apps ay hadlang sa akin.)				

24. I avoid online food delivery due to hidden charges (service fees, tax, etc.). (Iniwasan ko ang online food delivery dahil sa kalakip na mga singil, tulad ng service fees, tax, etc.)				
25. I would use online food delivery more frequently if it were more affordable. (Gagamitin ko ang online food delivery nang mas madalas kung ito ay mas abot-kaya.)				
Digital Barriers & Technical Issues				
26. I find food delivery apps difficult to use due to lack of digital skills or complexity. (Nahirapan akong gamitin ang mga food delivery apps dahil sa kakulangan ng mga digital na kasanayan o pagiging kumplikado nito.)				
27. My lack of access to a smartphone or stable internet prevents me from ordering food online. (Ang kawalan ko ng access sa isang smartphone o stable internet ay pumipigil sa akin na mag-order ng pagkain online.)				
28. I sometimes face payment issues when ordering food online. (Minsan ay nahaharap ako sa mga isyu sa pagbabayad kapag nag-o-order ng pagkain online.)				
29. The lack of customer support discourages me from using food delivery platforms. (Ang kakulangan sa customer support ang pumipigil sa akin gumamit ng food delivery platforms.)				
Service & Reliability Concerns				
30. I have had negative experiences with late deliveries. (May negatibong karanasan ako sa tagal ng pag-deliver.)				
31. I avoid food delivery due to past issues with incorrect or missing orders. (Iniwasan ko ang food delivery dahil sa mga nakaraang isyu sa hindi tama o nawawalang mga order.)				
32. I don't trust food delivery platforms to handle my food safely. (Hindi ako nagtitiwala sa mga food delivery platform upang ligtas na pangasiwaan ang aking pagkain.)				
33. Limited food choices in food delivery apps prevent me from using them. (Ang mga limitadong pagpipilian ng pagkain sa mga food delivery apps ay pumipigil sa akin na gamitin ang mga ito.)				
34. I avoid online food delivery because of concerns about food freshness. (Iniwasan ko ang online food delivery dahil sa mga alalahanin tungkol sa kalidad ng pagkain.)				
Trust & Security Concerns				
35. I avoid food delivery due to concerns about data security or fraud. (Iniwasan ko ang food delivery dahil sa mga alalahanin tungkol sa seguridad ng data o panloloko.)				
36. I do not feel comfortable using cashless payment options for food delivery. (Hindi ako kumportable sa paggamit ng mga opsyon cashless payment sa food delivery.)				
37. I worry that delivery personnel might mishandle my order. (Nag-aalala ako na baka mamali ng paghawak ang mga delivery personnel sa order ko.)				
38. I prefer buying food in-person rather than ordering online. (Mas gusto kong bumili ng pagkain nang personal kaysa mag-order online.)				

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

123

Appendix C

Letter to the Validators

March 6, 2025

Dr. Lea Nisperos
Program Head
Department of Information Technology

Dear Dr. Nisperos,

We hope this letter finds you well. We are conducting a study titled **"Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City."** As part of our research process, we seek your expertise in validating our research instrument to ensure its accuracy, reliability, and relevance.

Your insights and evaluation will help refine our survey questions, rating scales, and overall methodology to align with research standards and objectives. We kindly ask for your feedback regarding the items' clarity, coherence, and appropriateness.

Rest assured that your contributions will be acknowledged in our study, and we highly appreciate your time and effort in this validation process. Should you have any questions or require further information, please contact us at the institution email below.

Thank you for your valuable support and expertise.

Sincerely,



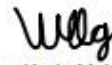
Arvin H. Candelaria
2022-109838@rtu.edu.ph



Vince Saiha T. Arcos
2022-108920@rtu.edu.ph



Rosalyn N. Abalos
2022-109939@rtu.edu.ph

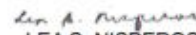


Dan Karlo M. Vullag
2018-107163@rtu.edu.ph

Noted by:



Faustino E. Oguan Jr. PhD
Thesis Professor


LEA S. NISPEROS

Signature Over Printed Name of the Validator

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

124

March 21, 2025

Mr. Hans Christian F. Delos Reyes, Rpm, CHRA
Regional Well-Being Coordinator
Concentrix, Philippines

Dear Mr. Delos Reyes,

We hope this letter finds you well. We are conducting a study titled "**Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City.**" As part of our research process, we seek your expertise in validating our research instrument to ensure its accuracy, reliability, and relevance.

Your insights and evaluation will help refine our survey questions, rating scales, and overall methodology to align with research standards and objectives. We kindly ask for your feedback regarding the items' clarity, coherence, and appropriateness.

Rest assured that your contributions will be acknowledged in our study, and we highly appreciate your time and effort in this validation process. Should you have any questions or require further information, please feel free to contact us at the institution email below.

Thank you for your valuable support and expertise.

Sincerely,


Arvin H. Candelaria
2022-109838@rtu.edu.ph


Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Vince Saina T. Arcos
2022-108920@rtu.edu.ph


Dan Karlo M. Vullag
2022-109939@rtu.edu.ph

Noted by:


Faustino E. Oguan Jr. PhD
Thesis Professor


03/21/25
Mr. Hans Christian F. Delos Reyes
Signature Over Printed Name of the Validator

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

125

March 25, 2025

Dr. Omar R. Manlapas
Program Head
Department of Political Science

Dear Dr. Manlapas,

We hope this letter finds you well. We are conducting a study titled **"Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City."** As part of our research process, we seek your expertise in validating our research instrument to ensure its accuracy, reliability, and relevance.

Your insights and evaluation will help refine our survey questions, rating scales, and overall methodology to align with research standards and objectives. We kindly ask for your feedback regarding the items' clarity, coherence, and appropriateness.

Rest assured that your contributions will be acknowledged in our study, and we highly appreciate your time and effort in this validation process. Should you have any questions or require further information, please contact us at the institution email below.

Thank you for your valuable support and expertise.

Sincerely,


Arvin H. Candelaria
2022-109838@rtu.edu.ph


Vince Salina T. Arcos
2022-108920@rtu.edu.ph


Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Dan Karlo M. Vullag
2018-107163@rtu.edu.ph

Noted by:

Faustino E. Oguan Jr. PhD
Thesis Professor


DR. OMAR R. MANLAPAS
Signature Over Printed Name of the Validator

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

126

April 1, 2025

Dr. Elvira Cabadsan-Labartine

Associate Professor IV, Department of Food Technology
College of Science
Polytechnic University of the Philippines

Dear Dr. Labartine,

We hope this letter finds you well. We are conducting a study titled **"Digital Platforms and Household Food Security: Understanding Access, Effectiveness, and Barriers in Mandaluyong City."** As part of our research process, we seek your expertise in validating our research instrument to ensure its accuracy, reliability, and relevance.

Your insights and evaluation will help refine our survey questions, rating scales, and overall methodology to align with research standards and objectives. We kindly ask for your feedback regarding the items' clarity, coherence, and appropriateness.

Rest assured that your contributions will be acknowledged in our study, and we highly appreciate your time and effort in this validation process. Should you have any questions or require further information, please contact us at the institution email below.

Thank you for your valuable support and expertise.

Sincerely,


Arvin B. Candelaria
2022-109838@rtu.edu.ph

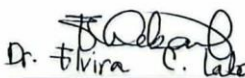

Vince Salita T. Arcos
2022-108920@rtu.edu.ph


Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Dan Karlo M. Vullag
2018-107163@rtu.edu.ph

Noted by:


Faustino E. Oguan Jr. PhD
Thesis Professor


Dr. Elvira C. Labartine
Signature Over Printed Name of the Validator

Appendix D

Letter to the Barangays

Dear Honorable Barangay Captain,

Greetings! We hope this letter finds you well

We are third-year students from Rizal Technological University currently pursuing a Bachelor of Science in Statistics. As part of our academic requirements, we are conducting a thesis entitled "Digital Platforms and Household Food Security." The purpose of our study is to explore how online food delivery services impact food security and accessibility.

In line with this, we humbly seek your permission to conduct a survey within your Barangay. The survey will involve a set of questionnaires to be answered by selected residents and will be conducted with utmost respect for their privacy and confidentiality. Each interview is estimated to take approximately 10 minutes.

We believe that the insights gathered from your barangay will be instrumental in shaping our research findings and contributing to broader discussions on combating misinformation in the digital age. Rest assured, all results will be shared with your barangay for transparency and potential community benefit.

We would be happy to discuss further details at your convenience and ensure that our team adheres to all protocols required by your barangay during the survey process.

Thank you very much for considering our request. We look forward to your positive response and are deeply grateful for your support in this academic endeavor.

Sincerely,


Arvin H. Gandelaria
2022-109838@rtu.edu.ph

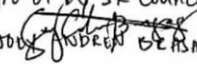

Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Vince Salha T. Arcos
2022-108920@rtu.edu.ph


Dan Karlo M. Vullag
2022-109939@rtu.edu.ph

Noted by:

Faustino E. Oguan Jr. PhD
Thesis Professor

care of by SK Council

JOHN ANDREW BEASMA

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

128

Dear Honorable Barangay Captain,

Greetings! We hope this letter finds you well

We are third-year students from Rizal Technological University currently pursuing a Bachelor of Science in Statistics. As part of our academic requirements, we are conducting a thesis entitled "Digital Platforms and Household Food Security." The purpose of our study is to explore how online food delivery services impact food security and accessibility.

In line with this, we humbly seek your permission to conduct a survey within your Barangay. The survey will involve a set of questionnaires to be answered by selected residents and will be conducted with utmost respect for their privacy and confidentiality. Each interview is estimated to take approximately 10 minutes.

We believe that the insights gathered from your barangay will be instrumental in shaping our research findings and contributing to broader discussions on combating misinformation in the digital age. Rest assured, all results will be shared with your barangay for transparency and potential community benefit.

We would be happy to discuss further details at your convenience and ensure that our team adheres to all protocols required by your barangay during the survey process.

Thank you very much for considering our request. We look forward to your positive response and are deeply grateful for your support in this academic endeavor.

Sincerely,


Arvin H. Candelaria
2022-109838@rtu.edu.ph


Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Vince Salha T. Arcos
2022-108920@rtu.edu.ph


Dan Karlo M. Vullag
2022-109939@rtu.edu.ph

Noted by:

Faustino E. Oguan Jr. PhD
Thesis Professor

Received by: 
Kag. 
4/22/25 9:22pm

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

129

Dear Honorable Barangay Captain,

Greetings! We hope this letter finds you well

We are third-year students from Rizal Technological University currently pursuing a Bachelor of Science in Statistics. As part of our academic requirements, we are conducting a thesis entitled "Digital Platforms and Household Food Security." The purpose of our study is to explore how online food delivery services impact food security and accessibility.

In line with this, we humbly seek your permission to conduct a survey within your Barangay. The survey will involve a set of questionnaires to be answered by selected residents and will be conducted with utmost respect for their privacy and confidentiality. Each interview is estimated to take approximately 10 minutes.

We believe that the insights gathered from your barangay will be instrumental in shaping our research findings and contributing to broader discussions on combating misinformation in the digital age. Rest assured, all results will be shared with your barangay for transparency and potential community benefit.

We would be happy to discuss further details at your convenience and ensure that our team adheres to all protocols required by your barangay during the survey process.

Thank you very much for considering our request. We look forward to your positive response and are deeply grateful for your support in this academic endeavor.

Sincerely,


Arvin H. Candelaria
2022-109838@rtu.edu.ph


Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Vince Saiha T. Arcos
2022-108920@rtu.edu.ph


Dan Karlo M. Vullag
2022-109939@rtu.edu.ph

Noted by:

Faustino E. Oguan Jr. PhD
Thesis Professor

RECEIVED
APR 22 2025
BY: Julius 4:14 PM

RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

130

Dear Honorable Barangay Captain,

Greetings! We hope this letter finds you well

We are third-year students from Rizal Technological University currently pursuing a Bachelor of Science in Statistics. As part of our academic requirements, we are conducting a thesis entitled "Digital Platforms and Household Food Security." The purpose of our study is to explore how online food delivery services impact food security and accessibility.

In line with this, we humbly seek your permission to conduct a survey within your Barangay. The survey will involve a set of questionnaires to be answered by selected residents and will be conducted with utmost respect for their privacy and confidentiality. Each interview is estimated to take approximately 10 minutes.

We believe that the insights gathered from your barangay will be instrumental in shaping our research findings and contributing to broader discussions on combating misinformation in the digital age. Rest assured, all results will be shared with your barangay for transparency and potential community benefit.

We would be happy to discuss further details at your convenience and ensure that our team adheres to all protocols required by your barangay during the survey process.

Thank you very much for considering our request. We look forward to your positive response and are deeply grateful for your support in this academic endeavor.

Sincerely,


Arvin M. Candelaria
2022-109838@rtu.edu.ph


Rosalyn N. Abalos
2022-109939@rtu.edu.ph


Vince Salina T. Arcos
2022-108920@rtu.edu.ph


Dan Karlo M. Vullag
2022-109939@rtu.edu.ph

Noted by:

Faustino E. Oguan Jr. PhD
Thesis Professor



RIZAL TECHNOLOGICAL UNIVERSITY

COLLEGE OF ARTS AND SCIENCES

131

Appendix E

Letter to the Grammarian

RTU-ACAD-FAC-F007

Control No. _____



RIZAL TECHNOLOGICAL UNIVERSITY
Cities of Mandaluyong and Pasig

☒ CAS

☐ CBET

☐ CED

☐ CEIT

☐ GS

☐ IPE

REQUEST FOR LANGUAGE AND FORMAT EDITING

July 7, 2025

Date

Thesis/Feasibility/Project Study Title:

DIGITAL FOOD PLATFORMS AND HOUSEHOLD FOOD SECURITY: UNDERSTANDING ACCESS, EFFECTIVENESS, AND BARRIERS

Number of Pages:

138

Researchers:

Arvin H. Candelaria

Rosalyn N. Abalos

Vince Saiha T. Arcos

Dan Karlo M. Vullag

Date Due:

July 9, 2025

I certify that I have edited the grammar and the format of the above-mentioned thesis/feasibility/project study and that all my corrections have been incorporated into the manuscript. It is now ready for bookbinding.

Prof. Gabriel C. Flores

Signature over Printed Name of Editor

July 8, 2025

Date

Copies to: Adviser, Editor, Researchers (to be included in the appendices)

Rizal Technological University

Revised 0

Effectivity April 2016

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	132
	CURRICULUM VITAE ARVIN HABANA CANDELARIA 2022-109838@rtu.edu.ph candelariaarvinh07@gmail.com Objective Dedicated working student pursuing a Bachelor's degree in Statistics at Rizal Technological University (Boni Campus) with hands-on experience in a BPO industry. Seeking to leverage analytical skills, adaptability, and a strong work ethic to contribute to data-driven decision-making while growing expertise in statistical analysis and problem-solving. Skills Good Written and Oral Communication Skills Problem-Solving Skills Time management skills Leadership skills Education • Mataas na Paaralang Neptali A. Gonzales 2020-2021 General Academic Strand (Senior High School) • Rizal Technological University - Boni Campus Bachelor of Science in Statistics Experience • Concentrix – WCC September 18, 2023 -November 20, 2023 Customer Service Adviser (SEASONAL) Handled Customers’ Initial Enrollment and Open Enrollment and customers complaints as a Benefit Administrator. • Sutherland Global Philippines Services Inc. July 22,2024 – September10, 2024 Associate-CS Internet Sports Streaming Venu chat support in the US, assisting customers with their subscriptions (bundled and standalone subscriptions). • Foundever Asia, Inc. October 4, 2024 - Present Date CSR Inbound 1 Being a customer service representative in the Auto loan financial industry, ALLY, currently in Retail. Answering multiple inbound inquiries from the customers in the US about their lien release status, payment assistance, online assistance, and contract changes or modifications from their dealership.	

CURRICULUM VITAE**VINCE SAIHA ARCOS**

2022-108920@rtu.edu.ph | saihaarcos@gmail.com

Objective

A motivated professional seeking a challenging role in a reputable organization to apply their skills, dedication, and eagerness to learn. Committed to contributing to team success through adaptability, efficiency, and a focus on growth.

Skills

Time Management

Teamwork

Adaptability

Leadership

Education

- Rizal Technological University

Bachelor of Science in Statistics

- Mataas na Paaralang Neptali A. Gonzales

2021-2022

(Science, Technology, Engineering, Mathematics) Senior High School

Experience

Warehouse Staff

November 21, 2022 - December 22, 2022

Repack products or parcels according to the correct specifications.

Label Parcels accurately based on the order list

Help organize and clean the warehouse or packing area.

CURRICULUM VITAE**ROSALYN ABALOS**

2022-109939@rtu.edu.ph | rosalynabalos07@gmail.com

Objective

Seeking a challenging role in a rapidly growing organization that provides diverse opportunities within a modern and dynamic business environment.

Skills

Eager to learn
Hard Working
Service and Support
Trustworthy and reliable
Strong Work Ethics

Education

- Rizal Technological University
BS. STATISTICS
- Arellano University Juan Sumulong Campus
2020-2022
(Humanities and Social Sciences) Senior high school

Experience

- Bigbrew Pureza September 14, 2022 – November 14, 2022
Cashier / Barista
Assisted customers by providing information on available drinks and helping them select their preferred flavors of milk tea and coffee.
Prepared and served a variety of drinks, including pearl milk tea and coffee.
Maintained cleanliness and organization of the work area in compliance with health and safety standards.

	RIZAL TECHNOLOGICAL UNIVERSITY	
	COLLEGE OF ARTS AND SCIENCES	135
	CURRICULUM VITAE DAN KARLO M. VULLAG 2018-107163@rtu.edu.ph Vullagkarlo@gmail.com Objective Motivated professional seeking an opportunity to learn directly from industry experts through hands-on training and real-world project exposure. Committed to applying and enhancing technical and interpersonal skills under experienced mentorship, to build a robust foundation for a long-term, growth-oriented career aligned with evolving industry demands. Skills Time management Emotional Intelligence Adaptability/Flexibility Education • Rizal Technological University – Boni Campus 2018- Present Date Bachelor of Science in Statistics • Philippine Christian University 2016-2018 Senior High School • Mariano Marcos Memorial High School 2012- 2016 High School • Margarita Roxas De Ayala Elementary School 2006- 2012 Experience Barista and Cashier 2021-2023	